CHAPTER 1

Introduction to Intellectual Property Law

CHAPTER OVERVIEW

Intellectual property law protects the results of human creative endeavor. Intellectual property is generally thought to comprise four separate fields of law: trademarks, copyrights, patents, and trade secrets. A *trademark* is a word, name, symbol, or device used to identify and distinguish one's goods or services and to indicate their source. Rights in trademarks are created by use of a mark; registration with the U.S. Patent and Trademark Office (USPTO) is not required, although it offers certain advantages. *Copyright* protects original works of authorship, including literary, musical, dramatic, artistic, and other works. Just as trademarks are protected from the moment of their first public use, copyright exists from the moment of creation of a work in fixed form; registration of a copyright with the U.S. Copyright Office, while affording certain benefits, is not required. A *patent* is a grant from the U.S. government that permits its owner to exclude others from making, selling, using, or importing an invention. Patents exist only upon issuance by the USPTO. A *trade secret* consists of any valuable commercial information that, if known by a competitor, would provide some benefit or advantage to the competitor. No registration or other formalities are required to create a trade secret, and trade secrets endure as long as reasonable efforts are made to protect their secrecy.

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INTELLECTUAL PROPERTY LAW BASICS

Intellectual Property Defined

There are three distinct types of property that individuals and companies can own: **real property** refers to land or real estate; **personal property** refers to specific items and things that can be identified, such as jewelry, cars, and artwork; and *intellectual property* refers to the fruits or product of human creativity, including literature, advertising slogans, songs, or new inventions. Thus, property that is the result of thought, namely, intellectual activity, is called **intellectual property (IP)**. In some foreign countries, intellectual property (especially patents and trademarks) is referred to as **industrial property**.

Many of the rights of ownership common to real and personal property are also common to intellectual property. Intellectual property can be bought, sold, and licensed. Similarly, it can be protected against theft or infringement by others. Nevertheless, there are some restrictions on use. For example, if you were to purchase the latest bestseller by John Grisham, you would be entitled to read the book, sell it to another, or give it away. You would not, however, be entitled to make photocopies of the book and then distribute and sell those copies to others. Those rights are retained by the author of the work and are protected by copyright law.

The Rationale for Protection of Intellectual Property

Intellectual property is a field of law that aims at protecting the knowledge created through human effort in order to stimulate and promote further creativity. Authors who write books and musicians who compose songs would be unlikely to engage in further creative effort unless they could realize profit from their endeavors. If their work could be misappropriated and sold by others, they would have no incentive to create further works. Pharmaceutical companies would not invest millions of dollars into research and development of new drugs unless they could be assured that their inventions would enable them to recover these costs and develop additional drugs. Thus, not only the creators of intellectual property but the public as well benefit from protecting intellectual property.

On the other hand, if the owner of intellectual property is given complete and perpetual rights to his or her invention or work, the owner would have a monopoly and be able to charge excessive prices for the invention or work, which would harm the public. Intellectual property law attempts to resolve these conflicting goals so that owners' rights to reap the rewards of their efforts are balanced against the public need for a competitive marketplace. Thus, for example, under federal law, a patent for a useful invention will last for only 20 years from the date an application for the patent is filed with the USPTO. After that period of time, the patent expires, and anyone is free to produce and sell the product.

TYPES OF INTELLECTUAL PROPERTY

The term *intellectual property* is usually thought of as comprising four separate, but often overlapping, legal fields: trademarks, copyrights, patents, and trade secrets. Although each of these areas will be discussed in detail in the chapters that follow, a brief introduction to each discipline is helpful. (See chart on inside front and back covers of text comparing and contrasting the various types of intellectual property.)

Trademarks and Service Marks

What Is Protectable.

A trademark or service mark is a word, name, symbol, or device used to indicate the source, quality, and ownership of a product or service. A **trademark** is used in the marketing

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of a product (such as REEBOK® for shoes), while a **service mark** typically identifies a service (such as STARBUCKS® for retail outlet services). A trademark or service mark identifies and distinguishes the products or services of one person from those of another.

In addition to words, trademarks can also consist of slogans (such as THE KING OF BEERS® for Budweiser beer), designs (such as the familiar "swoosh" that identifies Nike products), or sounds (such as the distinctive giggle of the Pillsbury Doughboy).

Trademarks provide guarantees of quality and consistency of the product or service they identify. Thus, upon encountering the golden arches that identify a McDonald's restaurant, consumers understand the "Big Mac" they purchase in Chicago will be the same quality as one purchased in Seattle.

Image not available due to copyright restrictions

Companies expend a great deal of time, effort, and money in establishing consumer recognition of and confidence in their marks. Yet not all words, phrases, or symbols are entitled to protection as trademarks. A chain of stores that sells clothing could not obtain a registered trademark for "Clothing Goods" inasmuch as the name is generic, yet GAP® is a nationally recognized mark for the retail sale of clothing. Marks may not be protectable if they are generic in nature or merely descriptive of the type of products or services they identify. Generally, marks that are protectable are those that are coined (such as KODAK®), arbitrary (such as SHELL® for gasoline), or suggestive (such as STAPLES® for office supplies).

Federal Registration of Trademarks. Interstate use of trademarks is governed by federal law, namely, the U.S. Trademark Act (also called the Lanham Act), found at 15 U.S.C. §§ 1051 *et seq*. See Appendix E. Additionally, trademarks are provided for in all 50 states so that marks that cannot be federally registered with the USPTO because they are not used in interstate commerce can be registered in the state in which they are used.

In the United States, trademarks are generally protected from their date of first public use. Registration of a mark is not required to secure protection for a mark, although it offers numerous advantages, such as allowing the registrant to bring an action in federal court for infringement of the mark. Applications for federal registration of trademarks are made with the USPTO. Registration is a fairly lengthy process, generally taking anywhere from 10 to 24 months or even longer. The filing fee is $325 per mark per class of goods or services covered by the mark if the application is filed electronically.

A trademark registration is valid for 10 years and may be renewed for additional 10-year periods thereafter as long as the mark is in use in interstate commerce. Additionally, registrants are required to file an affidavit with the USPTO between the fifth and sixth years after registration and every 10 years to verify the mark is in continued use. Marks not in use are then available to others.

Trademarks are among the most visible items of intellectual property, and it has been estimated that the average resident of the United States encounters approximately 1,500 different trademarks each day and 30,000 if one visits a supermarket. A properly selected, registered, and protected mark can be of great value to a company or individual desiring to establish and expand market share. There is perhaps no better way to maintain a strong position in the marketplace than to build goodwill and consumer recognition in the identity selected for products and services and then to protect that identity under federal trademark law.

Copyrights

What Is Protectable. **Copyright** is a form of protection governed exclusively by federal law (17 U.S.C. §§ 101 *et seq*.) granted to the authors of

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original works of authorship, including literary, dramatic, musical, artistic, and certain other works. See Appendix E. Thus, books, songs, plays, jewelry, movies, sculptures, paintings, and choreographic works are all protectable. Computer software is also protectable by copyright.

Copyright protection is available for more than merely serious works of fiction or art. Marketing materials, advertising copy, and cartoons are also protectable. Copyright is available for original works; no judgment is made about their literary or artistic quality. Nevertheless, certain works are not protectable by copyright, such as titles, names, short phrases, or lists of ingredients. Similarly, ideas, methods, and processes are not protectable by copyright, although the expression of those ideas is.

Copyright protection exists automatically from the time a work is created in fixed form. Thus, similar to trademark law, securing a registration for a work (with the U.S. Copyright Office) is not required for a work to be protected, although registration does provide significant advantages, such as establishing a public record of the copyright claim and providing a basis upon which an infringement suit may be brought in federal court and in which statutory damages and attorneys' fees may be recovered.

The owner of a copyright has the right to reproduce the work, prepare derivative works based on the original work (such as a sequel to the original), distribute copies of the work, and to perform and display the work. Generally, violations of such rights are protectable by infringement actions. Nevertheless, some uses of copyrighted works are considered "fair use" and do not constitute infringement, such as use of an insignificant portion of a work for noncommercial purposes or parody of a copyrighted work.

Federal Registration of Copyrights.

Neither publication nor registration of a work is required for copyright protection, inasmuch as works are protected under federal copyright law from the time of their creation in a fixed form. Registration, however, is inexpensive, requiring only a $35 filing fee (for applications filed electronically), and the process is expeditious. In most cases, the Copyright Office processes electronically filed applications in about three months.

Generally, copyrighted works are automatically protected from the moment of their creation for a term generally enduring for the author's life plus an additional 70 years after the author's death. After that time, the work will fall into the public domain and may be reproduced, distributed, or performed by anyone. The policy underlying the long period of copyright protection is that it may take several years for a painting, book, or opera to achieve its true value, and, thus, authors should receive a length of protection that will enable the work to appreciate to its greatest extent.

Patents

What Is Protectable.

A **patent** is a grant from the U.S. government that permits its owner to prevent others from making, using, importing, or selling an invention. There are three types of patents: utility patents, which are the most common patents and which cover useful inventions and discoveries (such as the typewriter, the automobile, and genetically altered mice); design patents, which cover new, original, and ornamental designs for articles (such as furniture); and plant patents, which cover new and distinct asexually reproduced plant varieties (such as hybrid flowers or trees).

Patent protection is available only for useful, novel, and nonobvious inventions. Generally, patent law prohibits the patenting of an invention that is merely an insignificant addition to or minor alteration of something already known. Moreover, some items cannot be protected by patent, such as pure scientific principles.

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Federal Registration of Patents. Patents are governed exclusively by federal law (35 U.S.C. §§ 100 *et seq.*). See Appendix E. To obtain a patent, an inventor must file an application with the USPTO (the same agency that issues trademark registrations) that fully describes the invention. Patent prosecution is expensive, time-consuming, and complex. Costs can run into the thousands of dollars, and it generally takes about three years for the USPTO to issue a patent.

Patent protection exists for 20 years from the date of filing of an application for utility patents (assuming that certain fees are paid to maintain the patent in force) and plant patents and 14 years from the date of grant for design patents. After this period of time, the invention falls into the public domain and may be used by any person without permission.

Patents promote the public good in that patent protection incentivizes inventors. In return for fully describing the invention in the patent application, the inventor is granted an exclusive but limited period of time within which to exploit the invention. After the patent expires, any member of the public is free to use, manufacture, or sell the invention. Thus, patent law strikes a balance between the need to protect inventors and the need to allow public access to important discoveries.

Trade Secrets

What Is Protectable. A **trade secret** consists of any valuable business information that, if known by a competitor, would afford the competitor some benefit or advantage. There is no limit to the type of information that can be protected as trade secrets; recipes, marketing plans, financial projections, and methods of conducting business can all constitute trade secrets. There is no requirement that a trade secret be unique or complex; thus, even something as simple and nontechnical as a list of customers can qualify as a trade secret as long as it affords its owner a competitive advantage and is not common knowledge.

If trade secrets were not protectable, companies would have no incentive to invest time, money, and effort in research and development that ultimately benefits the public. Trade secret law thus promotes the development of new methods and processes of doing business in the marketplace.

Protection of Trade Secrets. Although trademarks, copyrights, and patents are all subject to extensive statutory schemes for their protection, application, and registration, there is no equivalent federal law system for trade secrets, and no formalities are required to obtain rights to trade secrets. Trade secrets are generally protectable under various state statutes and cases and by contractual agreements between parties. For example, employers often require employees to sign confidentiality agreements in which employees agree not to disclose proprietary information owned by the employer.

If properly protected, trade secrets may last forever. On the other hand, if companies fail to take reasonable measures to maintain the secrecy of the information, trade secret protection may be lost. Thus, disclosure of the information should be limited to those with a "need to know" it so as to perform their duties; confidential information should be kept in secure or restricted areas; and employees with access to proprietary information should sign **nondisclosure agreements**. If such measures are taken, a trade secret can be protected in perpetuity.

Another method by which companies protect valuable information is by requiring employees to sign agreements promising not to compete with the employer after leaving the job. Such covenants are

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strictly scrutinized by courts, but in most states (but not California where such covenants are invalid), if they are reasonable in regard to time, scope, and subject matter, they are enforceable.

Other Intellectual Property Rights

Although the most common types of intellectual property are trademarks, copyrights, patents, and trade secrets, other intellectual property rights exist and will be discussed in the chapters that follow. Some of these rights include semiconductor chip protection, plant variety protection, the right of publicity, and rights relating to unfair competition, including passing off, misappropriation, and false advertising.

Additionally, intellectual property rights often intersect and overlap. Thus, the formula for Coca-Cola is a trade secret, while the distinctive script in which the words COCA-COLA® are displayed is a trademark. Generally, computer programs may be protectable under copyright law, patent law, and as trade secrets, while the name for a computer program, such as WINDOWS®, qualifies for trademark protection. Jewelry may be protected both under copyright and design patent law. Legal practitioners in the field of intellectual property law must fully understand how the various types of intellectual property intersect so that clients can achieve the widest possible scope of protection. For example, although an item of jewelry can be protected as a design patent, securing a patent is complex and expensive. Moreover, a design patent lasts only 14 years from the date of grant of the patent. In contrast, securing copyright protection for the same article of jewelry is easy and inexpensive. More importantly, copyright protection endures during the life of the work's creator and for 70 years thereafter. Trade secrets that are properly protected can endure perpetually. Thus, intellectual property owners need to consider the complementary relationships among trademark, copyright, patent, and trade secrets law so as to obtain the broadest possible protection for their assets.

AGENCIES RESPONSIBLE FOR INTELLECTUAL PROPERTY REGISTRATION

U.S. Patent and Trademark Office

The agency charged with granting patents and registering trademarks is the **U.S. Patent and Trademark Office (USPTO)**, one of several bureaus or agencies within the U.S. Department of Commerce. The USPTO, founded more than 200 years ago, employs more than 10,000 employees and is presently located in several buildings at its campus in Alexandria, Virginia. Mailing addresses vary depending on whether the matter relates to patents or trademarks. The USPTO is physically located at 600 Dulany Street, Alexandria, Virginia 22314. Its website is http://www.uspto.gov. The USPTO website offers a wealth of information, including helpful information about trademarks and patents, fee schedules, forms, and the ability to search and apply for trademarks and patents. Since 1991, under the Omnibus Budget Reconciliation Act, the USPTO has operated in much the same way as a private business, providing valued products and services to customers in exchange for fees that are used to fully fund USPTO operations. It uses no taxpayer funds.

The USPTO is one of the busiest of all government agencies, and as individuals and companies continue to value the importance of intellectual property assets, greater demands are being made on the USPTO. For example, from 2005 to 2011, the number of trademark applications received by the USPTO increased by 23 percent, and the number of patent applications received increased 31 percent.

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In 2011, the USPTO issued 244,430 patents and registered 177,661 trademarks.

Legislation passed in 1997 established the USPTO as a performance-based organization that is managed by professionals, resulting in the creation of a new political position, Under Secretary of Commerce for Intellectual Property and Director of the USPTO. Changing the USPTO from a mere governmental agency to a governmental corporation made the USPTO equivalent to other similar organizations, such as the Tennessee Valley Authority and the Federal Deposit Insurance Corporation. Performance-based organizations have considerable flexibility in personnel matters and set specific goals and objectives to achieve. In brief, the USPTO operates more like a business with greater autonomy over its budget, hiring, and procurement. Additionally, the USPTO website's searchable database includes information about all U.S. patents from the first patent issued in 1790 to the most recent, with full information for all patents since 1976 and the text and images of more than four million pending and registered federal trademarks. Users can view, download, and print the images of these patents and trademarks. The USPTO has nearly completed its transition from paper to electronic filing for both trademarks and patents. Nearly 100 percent of trademark applications and about 90 percent of patent applications were filed electronically in 2011.

The USPTO is led by the Under Secretary of Commerce for Intellectual Property and Director of the U.S. Patent and Trademark Office (the "Director"), who is appointed by the president. The Secretary of Commerce appoints a Commissioner for patents and a Commissioner for trademarks.

Cases relating to IP law are published in a variety of sources. One excellent reporter is *United States Patent Quarterly* (U.S.P.Q.), covering IP cases (relating to patents, trademarks, copyrights, and trade secrets) from 1929 to 1986, and U.S.P.Q.2d, covering IP cases from 1987 to date. In addition to publishing various federal cases relating to patents and trademarks, this set, published by the Bureau of National Affairs (BNA), Inc., also publishes administrative decisions of the Commissioner of Patents and Trademarks. Subscribers to the set receive weekly advance sheets with the most current cases; bound volumes are issued quarterly. Most law firms that specialize in IP work subscribe to this set. The set is also available through LexisNexis and Westlaw, the computer-assisted legal research systems.

Additionally, numerous cases are available through the USPTO website.

Library of Congress

The **Library of Congress**, sometimes referred to as "Jefferson's Legacy," was established in 1800 as a legislative library. It is America's oldest, national cultural institution and is the largest library in the world. Thomas Jefferson is considered the founder of the Library of Congress, and his personal library is at the heart of the library, inasmuch as in 1814 the library's 3,000 volumes were burned by the British, and the next year Jefferson sold his personal library collection of 6,487 volumes to the Library of Congress for $23,950.

The U.S. Copyright Office has been a part of the Library of Congress since 1870 and is in charge of examining the approximately 600,000 copyright applications filed each year, issuing registrations, and maintaining copyright deposits in its vast collection.

The Copyright Office is located at 101 Independence Avenue SE, Washington, DC 20559–6000, and its website is http://www.copyright.gov. Basic information about copyrights, forms, and other valuable information can be obtained for free and downloaded from the Copyright Office's website.

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ETHICS EDGE

DUTY OF COMPETENCE

The duty of competent legal representation is so important that it is the first substantive statement promulgated by the Model Rules of Professional Conduct of the American Bar Association. Rule 1.1 provides that competent representation requires the legal knowledge, skills, thoroughness, and preparation necessary for the representation of a client. Thus, although only an attorney may give legal advice and counsel a client as to the best strategy to protect his or her intellectual property, paralegals are expected to provide competent assistance. Paralegals thus have a duty to keep current with IP developments and cases in order to fulfill this duty of competence.

INTERNATIONAL ORGANIZATIONS, AGENCIES, AND TREATIES

There are a number of international organizations and agencies that promote the use and protection of intellectual property. Although these organizations are discussed in more detail in the chapters to follow, a brief introduction may be helpful.

- **International Trademark Association (INTA)** is a not-for-profit international association composed chiefly of trademark owners and practitioners. More than 5,500 trademark owners and professionals in more than 190 countries belong to INTA, together with others interested in promoting trademarks. INTA offers a wide variety of educational seminars and publications, including many worthwhile materials available at no cost on the Internet (see INTA's home page at http://www.inta.org). INTA is located at 655 Third Avenue, 10th Floor, New York, NY 10017-5617 (212/642-1700).
- **World Intellectual Property Organization (WIPO)** was founded in 1970 and is a specialized agency of the United Nations whose purposes are to promote intellectual property throughout the world and to administer 24 treaties dealing with intellectual property, including the Paris Convention, Madrid Protocol, the Trademark Law Treaty, the Patent Cooperation Treaty, and the Berne Convention. More than 180 nations are members of WIPO. WIPO is headquartered in Geneva, Switzerland, and its home page is http://www.wipo.int
- **World Trade Organization (WTO)** was organized in 1995 and deals with rules of trade among its more than 150 member nations. It resolves trade disputes and administers various agreements, including those relating to intellectual property. It is headquartered in Geneva, Switzerland, and its website is http://www.wto.org

There are also a number of international agreements and treaties that affect intellectual property. Among them are the following:

- **Berne Convention for the Protection of Literary and Artistic Works (the Berne Convention).** The Berne Convention was created

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in 1886 under the leadership of Victor Hugo to protect literary and artistic works. It has more than 160 member nations. The United States became a party to the Berne Convention in 1989. The Berne Convention is administered by WIPO and is based on the precept that each member nation must treat nationals of other member countries like its own nationals for purposes of copyright (the principle of "national treatment").

- **Madrid Protocol.** The Madrid Protocol came into existence in 1996 and allows trademark protection for more than 80 countries, including all 27 countries of the European Union, by means of a centralized, trademark-filing procedure. The United States implemented the terms of the Protocol in late 2003. This treaty facilitates a one-stop, low-cost, efficient system for the international registration of trademarks by permitting a U.S. trademark owner to file for international registration in any number of member countries by filing a single, standardized application form with the USPTO, in English, with a single set of fees.
- **Paris Convention.** One of the first treaties or "conventions" designed to address trademark protection in foreign countries was the Paris Convention of 1883, adopted to facilitate international patent and trademark protection. The Paris Convention is based on the principle of reciprocity, so that foreign trademark and patent owners may obtain in a member country the same legal protection for their marks and patents as can citizens of those member countries. Perhaps the most significant benefit provided by the Paris Convention is that of priority. An applicant for a trademark has six months after filing an application in any of the more than 170 member nations to file a corresponding application in any of the other member countries of the Paris Convention and obtain the benefits of the first filing date. Similar priority is afforded for utility patent applications, although the priority period is one year rather than six months. The Paris Convention is administered by WIPO.
- **North American Free Trade Agreement (NAFTA).** The NAFTA came into effect on January 1, 1994, and is adhered to by the United States, Canada, and Mexico. The NAFTA resulted in some changes to U.S. trademark law, primarily with regard to marks that include geographical terms.
- **Agreement on Trade-Related Aspects of Intellectual Property (TRIPS).** Negotiated from 1986–1994, TRIPS is administered by the World Trade Organization and establishes minimum levels of protection that member countries must give to fellow WTO members. Computer programs must be protected as copyrightable literary works, and countries must prevent misuse of geographical names such as Roquefort or Champagne.

(See Appendix A, Table of Treaties.)

THE INCREASING IMPORTANCE OF INTELLECTUAL PROPERTY RIGHTS

Although people have always realized the importance of protecting intellectual property rights, the rapidly developing pace of technology has led to increased awareness of the importance of intellectual property assets. Some individuals and companies offer only knowledge. Thus, computer consultants, advertising agencies, Internet companies, and software implementers sell only brainpower. Similarly, some forms of intellectual property, such as domain names and moving images shown on a company's Web page, did not even exist until relatively recently.

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Internet domain names such as "www.ibm.com" are valuable assets that must be protected against infringement.

The International Intellectual Property Alliance estimates that total copyright industries accounted for 11 percent of the U.S. gross domestic product in 2007 and that more than 11 million workers are employed by these industries. Additionally, nearly $126 billion of U.S. exports now depend on some form of intellectual property protection, including pharmaceuticals, motor vehicles, and aircraft and associated equipment.

Moreover, the rapidity with which information can be communicated through the Internet has led to increasing challenges in the field of intellectual property. Within hours after the world premiere of the movie *Episode III—Revenge of the Sith*, counterfeit copies were available on the streets of New York City for just a few dollars, and the movie was also available on the website BitTorrent for free downloading. Books, movies, and songs can now be copied, infringed, and sold illegally with the touch of a keystroke.

The Office of the United States Trade Representative has estimated that U.S. industries lose between $200 billion and $250 billion annually from piracy, counterfeiting of goods, and other intellectual property infringements.

In many cases, the most valuable assets a company owns are its intellectual property assets. For example, the value of the trademarks and service marks owned by the Coca-Cola Company has been estimated at more than $70 billion, making it the world's most valuable brand. Thus, companies must act aggressively to protect these valuable assets from infringement or misuse by others. The field of intellectual property law aims to protect the value of such investments.

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- The value of Microsoft Corporation's brand is estimated to be $59 billion.
- In 2010, *Avatar* was the film illegally downloaded most often.
- In late 2009, Vice President Biden referred to copyright piracy as "flat, unadulterated theft."
- The USPTO collects more than $5 million each day in fees.
- In early 2011, Senator Charles Grassley stated that a recent report estimated that the global value of counterfeit and pirated goods exceeded $650 billion.
- Some of our U.S. presidents have been prolific inventors:
    - Thomas Jefferson invented the swivel chair, a macaroni machine, and a cipher wheel (used to code and decode messages). Jefferson did not apply for any patents, believing all people should have access to new technology.
    - George Washington registered a trademark for his brand of flour in 1772.
    - Abraham Lincoln, the only president to hold a patent, patented an invention in 1849 for a method of steering a boat through shallow waters. The model, made by Lincoln himself, is in the Smithsonian Institution. You may view the patent at the USPTO'S website by searching for Patent. No. 6,469.

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CHAPTER SUMMARY

The term *intellectual property* is generally thought of as comprising four overlapping fields of law: trademarks (protecting names, logos, symbols, and other devices indicating the quality and source of products and services); copyrights (protecting original works of authorship); patents (grants by the federal government allowing their owners to exclude others from making, using, or selling the owner's invention); and trade secrets (any commercial information that, if known by a competitor, would afford the competitor an advantage in the marketplace). Patents must be issued by the federal government, whereas rights in trademarks are created by use of marks, and rights in copyright exist from the time a work is created in fixed form. Nevertheless, registration of trademarks and copyrights offers certain advantages and benefits. Trade secrets are governed by various state laws, and registration is not required for existence and ownership of a trade secret. Trademarks and trade secrets can endure perpetually as long as they are protected, while copyrights and patents will fall into the public domain and be available for use by anyone after their terms expire.

As our world becomes increasingly reliant on technological advances, greater demands and challenges are made on IP practitioners. The field is an exciting and challenging one and offers significant opportunities for hands-on involvement by IP professionals.

CASE ILLUSTRATION

POLICIES UNDERLYING INTELLECTUAL PROPERTY LAW

Case: *Mazer v. Stein*, 347 U.S. 201 (1954)

Facts: After partners made china statuettes and obtained copyright registrations for them, other parties copied them and began using the statuettes as bases for table lamps. The original makers of the statuettes sued for copyright infringement.

Holding: The copyrights were valid and could be infringed. The economic philosophy underlying the clause empowering Congress to grant patents and copyrights is the conviction that encouragement of individual effort by personal gain is the best way to promote the public welfare through the talents of authors and inventors. The fact that the statuettes could be patented as lamps did not bar their protection under copyright law.

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CASE STUDY AND ACTIVITIES

Case Study: Your firm's client, Holiday Cruises, Inc., operates a cruise line. Its various ships make both domestic and international cruises. The cruise ships offer a wide variety of activities for their guests and provide on-board restaurants, offshore excursions, and many other entertainment options. When guests check in, they are issued a "Holi-Day Pass," a laminated card that allows them to purchase beverages and other items aboard ship. Guests also have access to Fit Ship, the fitness centers on board each ship. These fitness centers offer a wide variety of exercise classes and fitness machines and equipment, including a new type of resistance band created by one of Holiday's employees. Holiday advertises its cruises on television and in magazines, often using a song, "Holidays Ahead," a song composed by one of Holiday's employees. Holiday is currently considering offering cruises to various ports in Mexico and is conducting confidential market surveys to determine the level of interest in such cruises.

Activities: Identify the intellectual property Holiday might own.

ROLE OF PARALEGAL

Because of the increasing array of intellectual property (IP) that can be created in our high-tech society and the increasing ease with which it can be infringed, intellectual property law is a growing practice area. Ten years ago, few law firms had intellectual property law departments, and intellectual property matters were handled by small firms that specialized in the field. Today, nearly every large law firm has a department devoted exclusively to intellectual property, and IP professionals are courted and valued. In many cases, specialists in the field of intellectual property law are paid more than their counterparts in other fields. For example, the National Association of Legal Assistants reported in 2010 that paralegals who devote more than 40 percent of their time to IP matters make more than 12 percent more than the average paralegal.

Among the tasks commonly performed by IP paralegals are the following:

- Assisting in trademark searching to clear marks for use and preparing, filing, and monitoring trademark registration applications, maintenance, and renewal documents;
- Preparing, filing, and monitoring copyright registration applications;
- Assisting in patent searching and preparing, filing, and monitoring patent applications;
- Docketing all dates for responses to the USPTO and Copyright Office and docketing dates to maintain trademark and patent registrations in force;
- Serving as a liaison with clients and keeping them informed of all matters relating to their IP portfolios;
- Drafting license agreements for licensing of trademarks, copyrights, and patents;
- Preparing employment agreements and noncompetition agreements;
- Assisting in intellectual property audits to determine the extent and value of a client's intellectual property; and
- Assisting in protection of trade secrets by developing and implementing policies for protection of trade secrets.

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trademarks by permitting a trademark owner to file for registration in any or all Madrid Protocol member nations by filing one application with its home office, with one fee. The previous system required the filing of individual trademark applications in different languages in multiple trademark offices (with different renewal dates as a result), usually requiring retaining numerous attorneys. Moreover, managing one's international portfolio is far easer under the Madrid Protocol because there is only one document filed at the IB to renew all international registrations, and assignments can be recorded with effect for all designated countries by filing a single document with the IB.

Disadvantages of the Madrid Protocol.

There are few disadvantages of the Madrid Protocol. Some popular foreign filing countries, however, are not members of the Madrid Protocol, including Canada, Mexico, and most of Latin America. Another disadvantage is that if a mark registered under the Madrid Protocol is later assigned to a party in a nonmember country, the Madrid application or registration will be invalidated. Finally, there is no provision under the Madrid Protocol to revise or amend a registered mark.

The Madrid Agreement

Another treaty, the Madrid Agreement, similarly provides for international registration of trademarks among its 56 member nations. Countries may belong to either the Madrid Agreement, the Madrid Protocol, or both. The United States is not a member of the Madrid Agreement. The Madrid Protocol, although parallel to the Madrid Agreement, is considered more flexible and comprehensive. For example, the Madrid Protocol allows one to file an international application based on either a registration or an application in one's home country; the Madrid Agreement requires a registration—one cannot file an application for international registration based solely on an application. Moreover, the Madrid Agreement does not allow for transformation to a national registration of an international registration that has been invalidated due to central attack.

Developments in Eastern Europe

The disintegration of the Communist bloc into independent republics greatly changed trademark practices in Eastern Europe and the former Soviet Union. The Soviet Union, once a single nation, has now become 15 countries, each of which is attempting to achieve some degree of free market economic development. Many of these countries have their own trademark offices, and it is now possible to file applications in each country.

The breakup of Czechoslovakia led to the creation of two trademark offices, one in the Czech Republic and one in Slovakia. Each has adopted its own trademark laws, and each accepts trademark applications. It is also possible to file trademark applications in Croatia, Slovenia, and the former Yugoslav Republic of Macedonia. Some countries such as Iran are presently subject to U.S. trade embargoes, and U.S. citizens are not permitted to file applications there.

EFFECTS OF NEW INTERNATIONAL AGREEMENTS (NAFTA, TRIPS, AND THE TRADEMARK LAW TREATY)

The North American Free Trade Agreement (NAFTA) came into effect on January 1, 1994, and is adhered to by the United States, Canada, and Mexico. The most significant change in U.S. trademark law resulting from NAFTA is that trademarks that are primarily geographically deceptively misdescriptive cannot be registered in the United States, even if they have acquired secondary meaning.

The World Trade Organization (WTO) is located in Geneva and is a global international

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organization dealing with rules of trade among nations. Its primary function is to oversee and administer various agreements entered into by most of the world's trading nations. The agreements are created through a series of negotiations, called "rounds." The WTO was formally created by the Uruguay Round negotiations and was established in 1995, although it is a successor to the General Agreement on Tariffs and Trade (GATT), which itself was established after World War II. One hundred and fifty three countries are members of the WTO, and the United States has been a member since January 1, 1995. GATT was both an organization (which was replaced by the WTO) and an agreement (which still exists but has been amended and incorporated into TRIPS, discussed later).

The Uruguay Round negotiations brought intellectual property rights into the GATT-WTO system for the first time through an agreement called the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS). TRIPS is the most comprehensive multilateral agreement on intellectual property. The first of two key changes to U.S. trademark law resulting from TRIPS is that nonuse of a mark for three years must be shown for a registration to be canceled for such nonuse (prior to TRIPS, the United States followed a rule that two years of nonuse of a mark resulted in a presumption of abandonment). The second change is that TRIPS provides that all signatory countries must prevent the use of a trademark that misleads consumers as to the geographical origin of goods. Even stronger protection is provided for geographical indications for wines and spirits, in that TRIPS precludes registration of marks for wines and spirits unless they originate from the place named. For example, a wine bearing the mark SONOMA must originate in that region of the United States.

The **Trademark Law Treaty** was concluded in 1994 and implemented in the United States in 1998 under the Trademark Law Treaty Implementation Act. The purpose of the treaty is to simplify and streamline trademark registration procedures in various countries so that procedural requirements of different countries would be more consistent with each other. As a result of the treaty, the United States now only requires one specimen showing use of a mark (rather than three as was previously required) and allows abandoned applications to be revived if the applicant alleges unintentional delay (rather than unavoidable delay, a much stricter standard). The treaty is administered by WIPO, and nearly 50 countries are members.

INTERNATIONAL ASSOCIATIONS

There are a variety of international associations devoted to protecting the rights of trademark owners. The best known are the International Trademark Association (INTA) and WIPO.

International Trademark Association (INTA)

The International Trademark Association was founded in 1878 as the United States Trademark Association and is dedicated to the advancement and support of trademarks as valuable items of world commerce. It is a not-for-profit association that serves its members and actively pursues private and public policy matters concerning trademarks. More than 5,500 trademark owners and professionals from more than 190 countries belong to INTA, together with others interested in promoting trademarks such as law firms practicing in the field of trademarks, academics, and advertising agencies that deal with trademarks.

INTA has played a significant role in trademark legislation, including promotion of the passage of the U.S. Trademark Act in 1946, the 1988 Trademark Law Revision Act (designed to bring U.S. trademark law into conformity with that of

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that once Microsoft sold its products abroad, under the first sale doctrine the defendant was entitled to further resell them or do as it wished with the products.

Holding: The first sale defense can only be raised where claims involve domestically made copies of U.S. copyrighted goods. Because it was undisputed that the software imported by the defendant was manufactured in Ireland and first sold abroad, the defendant cannot assert the first sale defense. Because the goods were manufactured abroad rather than in the United States, the ruling in *Quality King* was inapplicable. Summary judgment was granted for Microsoft on its claim of copyright infringement.

CASE STUDY AND ACTIVITIES

Case Study: Holiday would like to seek copyright protection in several foreign countries for a book of photographs taken during various cruises. The book has been published in the United States and does not bear a copyright notice. The countries in which Holiday would like to seek protection are all members of the Berne Convention.

Activities: Discuss the protection that Holiday may be able to secure in these countries for its book of photographs.

ROLE OF PARALEGAL

Unless a client sells or distributes its products or works abroad, involvement by IP paralegals with international copyright issues will be minimal. Nevertheless, inasmuch as works can now be sold or distributed in foreign countries merely by the touch of a computer key, some familiarity with international copyright protection is needed. Intellectual property professionals will likely be involved in the following tasks:

- Monitoring issues related to international copyright by reading journals, articles, and other materials;
- Gathering information and publications from the U.S. Copyright Office related to international copyright protection, such as Circular 38a entitled "International Copyright Relations of the United States" and Circular 38b entitled "Highlights of Copyright Amendments Contained in the [Uruguay Round Agreements Act]";
- Retrieving the text of treaties to which the United States is a party either by locating them in conventional print form at law libraries or locating them on the Internet (the Berne Convention and the WIPO Internet treaties can be located through WIPO at http://www.wipo.int, and the Universal Copyright Convention can be located through the United Nations site at http://www.un.org or http://www.unesco.org); and
- Routinely monitoring various websites (see the resources at the end of this chapter and in Appendix C) related to intellectual property to keep abreast of new developments in the international arena.

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INTERNET RESOURCES

Federal laws relating to copyright:	http://www.law.cornell.edu http://www.gpo.gov/fdsys
Copyright Office:	http://www.copyright.gov (for Circulars 38a and 38b, relating to international copyright relations)
U.S. Trade Representative:	http://www.ustr.gov (for annual reports and information on efforts to protect intellectual property rights abroad; identification of foreign priority watch countries)
World Intellectual Property Organization:	http://www.wipo.int (for information on international copyright law and the text of the Berne Convention and the new WIPO Internet treaties)
UNESCO:	http://www.unesco.org (for text of Universal Copyright Convention)
World Trade Organization:	http://www.wto.org (for information on TRIPS, trade, and intellectual property issues)
International Intellectual Property Alliance:	http://www.iipa.com (for information on international protection of copyrighted materials)

DISCUSS QUESTIONS

1. Pablo Ortiz, a Spanish national, owned a copyright in a book in 1985 in the United States. The book, however, did not bear any copyright notices. Pablo timely filed a Notice of Intent to Enforce Rights under the URAA with the Copyright Office with regard to ABC Inc., which had begun publishing the book in the United States. What are the rights of the parties to the book?
2. Why are some copyright experts concerned about proposals to extend copyright protection to automatic databases that are the result of significant effort, time, and money?
3. Mel's copyrighted software is manufactured in the United States and then shipped to Scotland to be sold by XYZ Co. Mel has just discovered that Sam has imported the goods into the United States from Scotland (without Mel's permission). Mel has written Sam a letter informing Sam that he is infringing Mel's rights. Sam has responded that he has not infringed the copyright in the software because he has a right, under the first sale doctrine, to dispose of the software as he sees fit because once the goods were sold to XYZ, Mel's rights to distribution were exhausted. Discuss whether Sam's defense is likely to be upheld.
4. Using the facts in the previous question, would your answer be different if the goods were first manufactured and sold in Scotland and then imported into the United States by Sam? Would Sam's first sale defense be likely to be upheld? Discuss.

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5. Celia intends to file for copyright protection for her song in Argentina, which is a member of the Berne Convention. What does the Berne principle of "national treatment" mean for Celia with regard to copyright protection in Argentina?

USING INTERNET RESOURCES

1. Access the website of WIPO, and access information relating to the Berne Convention.
   a. When did Switzerland become a member of the Berne Convention?
   b. Briefly, what does Article 10 of the Berne Convention relate to or provide?
2. Access the website of WIPO, and access information relating to the WIPO Copyright Treaty. Review the "Summary" information. What two subject matters are to be protected by copyright under the Copyright Treaty?
3. Access the website of the Copyright Office.
   a. Search the records of the Copyright Office and locate document V8000P024. What is this document?
   b. Review Circular 38. What copyright relations do Iran and Egypt have with the United States?
4. Access the website of UNESCO and locate the Universal Copyright Convention. Briefly, what does Article II relate to or provide?

Go to http://www.paralegal.delmar.cengage.com for Quizzes, Forms, Chapter Resources, and additional information.

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PART FOUR

THE LAW OF PATENTS

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CHAPTER 17

Foundations of Patent Law

CHAPTER OVERVIEW

A patent is a legal right granted by the federal government that permits its owner to prevent others from making, using, selling, or importing an invention. There are three types of patents: utility patents, design patents, and plant patents. The great majority of patents are utility patents, granted for useful objects or processes. For more than 200 years, patents in the United States have been granted to the first to invent, assuming the invention or discovery is not known or used by others in the United States or patented or described in a printed publication in the United States or elsewhere. Effective March 16, 2013, however, and as a result of the Leahy-Smith America Invents Act of 2011, U.S. law will be harmonized with that of nearly all foreign countries so that patents will be awarded to the first to file the application.

Not all discoveries or inventions are eligible for utility patent protection. Patent protection is available only for a new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Thus, an inventor must demonstrate that the invention or discovery is useful, novel, and nonobvious. Generally, patent law prohibits the patenting of an invention that is merely an insignificant addition to or trivial alteration of something already in existence or already known.

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Some items are excluded from patent protection. For example, a mere arrangement of printed matter, naturally existing substances, some methods of doing business (which are mere abstract ideas), and scientific principles are unpatentable subject matter.

Applications for patents are filed with the U.S. Patent and Trademark Office (USPTO), and protection begins only when a patent is issued by the USPTO (as opposed to trademark protection, which arises from first use of a mark rather than registration with the USPTO, and copyright protection, which arises from creation of a work in a fixed form rather than from registration with the Copyright Office).

INTRODUCTION

The word *patent* is a shorthand expression for "letters patent." A **patent** is a grant from the U.S. government to exclude others from making, using, selling, or importing another person's new, nonobvious, and useful invention in the United States for the term of patent protection. After this period of exclusive protection (20 years from filing for utility and plant patents and 14 years from grant for design patents), the invention falls into the public domain and may be used by any person without permission. This right of exclusion is far different from the rights provided under copyright law. Under patent law, inventors can enjoin the making, using, selling, or importing of an infringing invention *even if it was independently created*. In contrast, copyright law protects only original works of authorship. If two writers independently compose the same poem, both are protected under copyright law. A patent allows its owner to exclude others from using the owner's invention; it does not provide any guarantee that its owner can use or sell the invention.

To obtain a patent, an inventor must file an application with the USPTO, the same agency of the Department of Commerce that issues trademark registrations. The application must describe the invention with specificity. The application will be reviewed by a USPTO examiner, and, if approved, the patent will issue. The patent is a written document that fully describes the invention.

Just as U.S. copyright law derives from the Constitution, so does patent law. The U.S. Constitution provides that Congress shall have the power "to promote the progress of science and useful arts, by securing for limited times to authors and inventors the exclusive right to their respective writings and discoveries." U.S. Const. art. I, § 8, cl. 8. The wording applicable to patents is *science, useful arts, inventors,* and *discoveries*.

Patents promote the public good in that patent protection incentivizes inventors. If inventors of useful discoveries could not protect their works from use or exploitation by others, there would be little motivation to expend effort, time, and money in creating inventions. The introduction of new products and processes benefits society. In return for the full disclosure to the public of the specifics of the invention, thus advancing science and technology, the inventor is given a limited period of time within which to exploit his or her invention and exclude others from doing so. Inventors are thus encouraged to create new products, and the public benefits from inventions that ultimately will fall into the public domain.

RIGHTS UNDER FEDERAL LAW

As stated, patent law derives from the Constitution. In 1790, pursuant to the direction provided in the Constitution, Congress passed the first patent statute, which in large part relied upon English law. Just three years later, the statute was replaced with a new act authored by Thomas Jefferson. These early

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acts provided the structural framework for U.S. patent law and specified the four basic conditions, still existing, that an invention must satisfy to secure patent protection:

1. The invention must be a utility, design, or plant patent.
2. It must be useful (or ornamental in the case of a design patent or distinctive in the case of a plant patent).
3. It must be novel in relation to the prior art in the field.
4. It must not be obvious to a person of ordinary skill in the field.

Revisions of federal patent statutes occurred in 1836 when the Patent Office was created and again in 1870 and 1897. Thereafter, in 1952, Congress enacted a new patent act, codified in Title 35 of the *United States Code*. The America Inventors Protection Act of 1999 (discussed in later chapters) also produced some major changes in patent law. In 2011, however, Congress passed the most significant revisions to federal patent law since 1952 when it enacted the **Leahy-Smith America Invents Act (AIA)**. Issues relating to patents are resolved solely by federal law. Moreover, development of patent law has evolved primarily through federal court decisions rather than the legislature. Just as seen in copyright law, where the term *writings* has been held to be broad enough to cover emerging technologies such as computer programs, the language in the 1793 act relating to the protectability of machines, manufactures, art (later changed to *process*), and compositions is broad enough to cover new developments such as computers and electronics. In 1982, Congress created a new court, the Court of Appeals for the Federal Circuit (CAFC), to exercise exclusive jurisdiction over all cases involving patent issues and to promote uniform interpretation of the U.S. patent statutes, which until then had been interpreted in often inconsistent ways by the various federal courts of appeals throughout the nation.

Unlike trademark and copyright law, both of which recognize common law rights in marks and works of authorship even without federal registration, patent law requires that an inventor secure issuance of a patent from the U.S. government to protect and enforce his or her rights against infringers. Inventors, however, may also secure some protection for their works under trade secret law (see Chapter 22). Moreover, some inventions, such as computer programs, are protectable under copyright law as well as patent law.

U.S. PATENT AND TRADEMARK OFFICE

Patents exist only by authority of government grant. The department of the government responsible for granting patents is the Department of Commerce, acting through the USPTO. The USPTO receives applications, reviews them, and issues or grants patents. The USPTO also publishes and disseminates patent information, records assignments of patents, maintains files of U.S. and foreign patents, and maintains a search room for public use in examining issued patents and records. The present address for mailing most patent-related documents is Mail Stop ___ (insert particular mail stop or box number per USPTO website instructions), Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313–1450. Because addresses may change, always check the USPTO website before submitting documents or correspondence to the USPTO. Most correspondence with the USPTO, however, is submitted electronically. Additionally, the USPTO website (http://www.uspto.gov) offers a wealth of general information, forms for downloading, patent statistics, news updates about issues affecting the USPTO and patent practice, schedules of patent fees, and other valuable information.

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As secretary of state, Thomas Jefferson was the first head of the Patent Office. Legend has it that the reason the files in which patents are kept and maintained are called *shoes* is that the first patent applications were stored in Jefferson's shoeboxes.

The practices and procedures relating to examination and issuance of patents are found in the USPTO publication ***Manual of Patent Examining Procedure*** **(MPEP)**, which most practitioners keep handy to serve as a reference tool for patent issues and questions. The entire text of the MPEP is available for viewing and downloading at the USPTO website.

Additionally, regulations relating to patents are found in Title 37 of the Code of Federal Regulations. These rules and regulations explain how the patent laws are to be implemented, provide procedures to be followed at the USPTO, and generally govern the day-to-day situations that may arise at the USPTO.

Various methods for locating patents are available. The University of New Hampshire's Franklin Pierce Center for Intellectual Property (http://ipmall.info) provides links to a variety of patent-related sites, including one listing famous patents, from Eli Whitney's cotton gin, to the application by Orville and Wilbur Wright for the airplane, to the first application for a computer program. Another site (http://www.colitz.com/site/wacky_new.html) identifies the "wacky patent of the month" and references issued patents for oddities such as a "pat on the back apparatus" and eye protectors for chickens.

Another way of locating and retrieving issued patents is through the USPTO patent database, which provides the full text of all patents issued since 1976 and full-page images of all patents issued since 1790. Patents from 1790 through 1975 are searchable only by patent number and the current patent classification system of identifying patents. Patents issued after 1975 are searchable through a variety of techniques, including patent number, inventor name, or topic (using Boolean connectors). The USPTO website offers search tips, and patent searching is discussed further in Chapter 18.

More than eight million patents have been issued since the first patent in 1790. The patent office is increasingly busy. In 2011, 536,604 patent applications were filed, and 244,430 patents were issued. Many experts attribute the onslaught of filings to applications for software, telecommunications inventions, and Internet-related inventions.

(See Exhibit 17–1 for a list of famous patents issued in the United States.)

PATENTABILITY

An invention must satisfy four basic requirements to be eligible for patent protection:

1. The invention must be one of the types specified by statute as patentable subject matter (namely, a utility, design, or plant patent).
2. The invention must be useful (if the application is for a utility patent).
3. The invention must be novel.
4. The invention must be nonobvious.

Patentable Subject Matter—Utility Patents

There are three distinct types of patents: utility patents, design patents, and plant patents. Utility patents are the most common and cover a wide variety of inventions and discoveries, including the typewriter, the automobile, the sewing machine, the zipper, the helicopter, sulfa drugs, gene sequences, and genetically altered mice. Design patents cover new, original, and ornamental designs for useful articles such as furniture, jewelry, and containers. Plant patents cover new and distinct asexually reproduced plant varieties, such as hybrid flowers or

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Patent Number	Inventor	Description	Date of Issuance
None	Samuel Hopkins	Improved potash process (first patent)	7/31/1790
None	Eli Whitney	Cotton gin	3/14/1794
None	Samuel Colt	Revolver	2/25/1836
1,647	Samuel F.B. Morse	Telegraph	6/20/1840
3,630	Linus Yale	Door lock	6/13/1844
6,281	Walter Hunt	Safety pin	4/10/1849
6,469	Abraham Lincoln	Buoying boats over shoals	5/22/1849
13,661	Isaac Singer	Sewing machine	10/9/1855
22,186	John Mason	Mason jar	11/30/1858
127,568	Richard Chesebrough	VASELINE® petroleum jelly	6/4/1872
140,245	Samuel Clemens	Scrapbook	6/24/1873
174,465	Alexander Graham Bell	Telephone	3/7/1876
182,346	Melville Bissell	Carpet sweeper	9/19/1876
223,898	Thomas Edison	Electric light	1/27/1880
388,850	George Eastman	Roll film camera	9/4/1888
473,653	Sarah Boone	Ironing board for sleeves	4/26/1892
504,038	Whitcomb Judson	Zipper	8/29/1893
644,077	Felix Hoffman	Aspirin	2/27/1900
686,046	Henry Ford	Automobile	11/5/1901
775,135	King Gillette	GILLETTE® safety razor	11/15/1904
821,393	Wilbur and Orville Wright	Airplane	5/22/1906
1,102,653	Robert Goddard	Two-stage rocket	7/7/1914
1,242,872	Clarence Saunders	Supermarket	10/9/1917
1,370,316	Harry Houdini	Diver's suit	3/1/1921
2,071,250	Wallace Carothers	Nylon	2/16/1937
2,177,627	Richard Drew	SCOTCH® cellophane tape	10/31/1939
2,682,235	Buckminster Fuller	Geodesic dome	6/29/1954
2,717,437	George de Mestrel	VELCRO® faster tape	9/13/1955
2,799,619	Seifter, Monaco, & Hoover	Tranquilizer	7/16/1957
4,270,182	Satya Asija	First computer program (software)	5/26/1981

EXHIBIT 17–1 Famous U.S. Patents

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trees. Because the vast majority of patents are utility patents, they will be discussed first. Design and plant patents will be discussed later in this chapter.

Federal law (35 U.S.C. § 101) establishes the subject matter that can be protected by a **utility patent**: "Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor . . . ."

Although ideas are not patentable, processes are. A **process** is a method of doing something to produce a given result. According to *Cochrane v. Deener*, 94 U.S. 780, 788 (1877), a process is "an act, or series of acts, performed upon the subject-matter to be transformed and reduced to a different state or thing. If new and useful, it is just as patentable as is a piece of machinery." A patent may be issued for a newly developed process or a new use of an already known process. Some examples of patented processes are the process for chrome plating; the process for making synthetic diamonds; processes for vulcanizing India rubber, smelting ore, tanning, and dyeing; Louis Pasteur's 1873 patented yeast process; and Clarence Birdseye's process for packaging frozen food. In fact, the first patent issued in the United States, to Samuel Hopkins in 1790, was entitled "improved potash process." In some instances, not only the process but also the result of the process is patentable, such as U.S. Patent No. 6,881,428 for the "process of making a lactose-free milk and the milk so processed," which combines a process (the method of manufacturing) with a product (the milk).

The other types of utility patents (machines, manufactures, and compositions of matter) are all products or items. A **machine** is a device with moving parts that accomplishes a result, such as a sewing machine or a blender. A **manufacture** includes anything under the sun that is made by humans; the term is broad enough to include a pitchfork and the once-patented Monopoly game. It is often said that the category of "manufactures" is a type of catchall category to encompass devices or items not easily classified as machines, compositions of matter, or processes. A **composition of matter** is a combination of two or more chemical or other substances into a product, such as a synthetic diamond or the fabric known as polyester. An invention can also consist of some new use of a process, machine, manufacture, or composition of matter as long as that new use demonstrates significant change from the original invention. Thus, new and nonobvious uses of old inventions and compositions are patentable.

Usefulness

The Constitution itself provides that patent protection is available for "*useful* arts," and 35 U.S.C. § 101, in defining what is patentable, states that patents are available for *useful* processes, machines, manufactures, and compositions of matter. Although **usefulness** is not a stringent standard, the invention must be of some benefit to society to be "useful." Mere novelties or inventions that conflict with scientific principles, such as a perpetual motion machine, are not patentable because they are not useful. In brief, such an invention has no usefulness because it does not work. Similarly, inventions whose only purpose is detrimental or fraudulent or would promote illegal ends cannot be patented inasmuch as, by definition, they are not useful. For example, a patent was denied for a process of making a low-cost tobacco leaf that resembled a more select leaf on the basis that its only purpose was to deceive consumers. *Rickard v. Du Bon*, 103 F. 868 (2d Cir. 1900). Inventions that serve to amuse or entertain are considered useful. Generally, a small degree of utility is sufficient to show that an invention satisfies the requirement of usefulness. In fact, the commercial success of an invention is evidence of its utility.

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The usefulness required of an invention must be present usefulness (determined at the time of invention), not usefulness purely for research purposes. Thus, patent protection will be denied to a drug whose usefulness cannot yet be shown or to a process, the result of which produces an article that has no current use. The fact that a drug, invention, or result of a process might show some benefit or usefulness at some time in the future is generally not sufficient and a patent therefor will be denied. Nevertheless, even if an invention or discovery shows no benefit to humans, it will be protected by patent law if usefulness for animals can be shown. Thus, drugs or compounds whose effectiveness has been demonstrated for animals can be patented even though their usefulness for humans cannot yet be shown. *In re Brana*, 51 F.3d 1560 (Fed. Cir. 1995). Applications that claim some drug or other substance is useful for humans must be accompanied by supporting evidence, usually results of tests or trials, and must show the substance is reasonably safe. To be patentable, an invention need not meet the stricter standards of regulatory agencies such as the Food and Drug Administration; it need only be "reasonably safe."

To ensure that the invention is useful, the application must disclose or specify the usefulness of the invention. To allow a patent that does not specify its utility would be to grant a patent on an entire range of unknown applications, thereby allowing an inventor to obtain a monopoly on an entire field of knowledge. Thus, patent applications must describe their specific advantage or usefulness so the public can benefit from the invention. Similarly, a patent will be denied when an invention fails to operate as described or claimed in the application.

Novelty

Section 101 of the Patent Act requires that an invention (or any improvement to an invention) be "new" or novel.

Introduction.

Current Section 102 of the Patent Act (effective only until March 16, 2013) elaborates on the requirement of **novelty** by setting forth certain situations that demonstrate the invention is not novel. Most codify the principle that will be recognized until March 16, 2013, that the first to invent will be granted a patent. If an invention is known or used by others, is the subject of an existing patent, or has been described or sold, then it is not novel and, accordingly, is not eligible for patent protection.

In such a case, it is deemed that the applicant's invention was anticipated and an application will be denied. Often called the **doctrine of anticipation**, this principle is intended to ensure that a second or junior inventor does not secure a monopoly on an invention that a senior inventor owns or that is in the public domain.

Effective March 16, 2013, the AIA creates a new **first to file system**. Every industrialized nation other than the United States uses a "first to file" patent priority system. In such a system, when more than one application claiming the same invention is filed, priority is given to the earlier-filed application. The United States, by contrast, has used a **first to invent system**, in which, if there is a conflict over priority of inventorship, priority is established through a complex and expensive proceeding (called an "interference proceeding") to determine which applicant invented the claimed invention first. In a first to file system, the filing date of the application is critical. This "line in the sand" filing date provides an objective and easy way of determining which inventor is entitled to a patent.

Because the U.S. "first to invent" system was out of step with that of nearly all other countries and required patent practitioners who filed applications in other countries to comply with two different systems, in 2011, Congress passed the AIA to move the United States to a first to file system. The new system, however, will continue to provide inventors the one-year **grace period** or statute of limitations they

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have always enjoyed: Applicants' own publication or disclosure of their invention that occurs within one year prior to filing will not act as prior art against their applications so as to bar the application. Similarly, disclosure by others during that time based on information obtained (directly or indirectly) from the inventor will not constitute prior art. This one-year grace period should continue to give U.S. applicants the time they need to prepare and file their applications. Because it is presumed that inventors will need some period of time to adjust to this entirely new system, the U.S. will follow its "first to invent" system until March 16, 2013, on which date it will convert to a "first to file" system with the one-year grace period described.

Applications and Novelty after March 16, 2013. Pursuant to 35 U.S.C. § 102, effective March 16, 2013, a person will be entitled to a patent unless the claimed invention was already patented, described in a printed publication, or in public use, on sale, or otherwise available to the public *anywhere in the world* before the filing date of the application for the invention. Nevertheless, as described previously, a disclosure made one year or less before the filing date will not bar the application as prior art if the disclosure was made by the inventor or by another who obtained the information disclosed directly or indirectly from the inventor. Thus, once the inventor publishes or discloses his or her invention, he or she has one year to file the patent application. Failure to file an application within the one-year grace period will bar issuance of a patent. After March 16, 2013, any public use or sale of the invention prior to the application filing date will preclude the granting of a patent unless the disclosure is by the inventor or one who obtained the subject matter directly or indirectly from the inventor.

Applications and Novelty until March 16, 2013. Until March 16, 2013, when the AIA converts the United States to a first to file system, a number of conditions will defeat novelty required under 35 U.S.C. § 102. Most of current Section 102's provisions are all intended to ensure patents are granted to the first inventor. The second or junior inventor's product or process is not novel in such circumstances. The conditions that will defeat novelty required under 35 U.S.C. § 102 and prevent the granting of a patent are the following:

- ***Section 102(a): Invention is known or used by others.*** If the invention is known or used by others in the United States or patented or described in a printed publication in this or a foreign country before its invention by the applicant, it cannot be patented. This provision articulates the "first to invent" rule that will be followed in the United States until March 16, 2013, and is directed to the acts of others. If another has used the invention, patented it, or described it in a printed publication before the applicant invented it, the invention is not novel, and an applicant who is not the first to invent is not entitled to receive a patent for the invention.
- ***Section 102(b): Invention is in use or on sale.*** If the invention is patented or described in a printed publication in this or a foreign country or in public use or on sale in this country more than one year before the application for the patent is filed, it cannot be patented. These requirements of current Section 102(b), typically called the **on sale bar**, are intended to ensure that inventors act promptly to secure protection for their inventions. Once an invention is in public use or offered for sale or sold in the United States or is patented or described in a printed publication anywhere, the inventor has a one-year grace period to file an application.

  The one-year grace period is often triggered by the inventor's own acts, such as offering the article for sale, describing it in a promotional literature, and so forth.

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The on sale bar of current 35 U.S.C. § 102(b) engenders many refusals by the USPTO. It is intended to encourage prompt action by inventors. It would be unfair to allow an inventor to use an invention for profit and delay filing for an application because such a delay has the effect of adding time to the term of protection for the patent and delaying its entry into the public domain.

Recall that the focus of Section 102(a) is on the acts of those *other than* the inventor. The focus of Section 102(b) is on the actions of both the inventor and others more than one year before the inventor files his or her patent application (although it is often the inventor's own acts that trigger the one-year time bar). Another distinction between the two subsections is that Section 102(a) bars a patent if the invention was patented or described in a printed publication anywhere in the world before the *invention* by the applicant; Section 102(b) bars a patent if the invention was printed or described anywhere in the world more than one year before an *application* is filed for the invention.

- ***Section 102(c): Invention has been abandoned.*** A person cannot obtain a patent if he or she has abandoned the invention. Abandonment is shown when the inventor expressly or impliedly demonstrates an intent to abandon his or her right to a patent. Abandonment usually requires an intentional act. Delay alone in filing an application does not usually constitute abandonment (although if no steps are taken to patent the invention and delay is unreasonable, abandonment may be found).
- ***Section 102(d): Invention is the subject of a foreign patent.***
- ***Section 102(e): Invention is described in a prior published application or patent.***
- ***Section 102(f): Inventor did not invent the invention.*** Only the inventor is entitled to a patent for the invention or discovery. The inventor is the person who conceived the specific invention. If the inventor derived the invention from another, a rejection under Section 102(f) is proper. Parties who work jointly on an invention may be joint inventors even though their contributions are not equal or simultaneous. An application for the resulting invention must name all inventors.
- ***Section 102(g): Invention was first invented by others.*** In determining priority of invention, the current statute provides that the dates the inventors conceived of the invention, the dates they reduced their inventions to practice, and their reasonable diligence in reducing the invention to practice will be considered. As discussed in Chapter 19, reduction to practice can be actual (making or building the invention or a prototype) or constructive (filing an application for a patent after invention).

Under the AIA, and effective March 16, 2013, all of these complex and subjective concepts related to determining who was the first to invent will be eliminated because the first to file system provides an objective and clear manner of determining novelty: The filing date controls.

Publication of Invention. Both before and after March 16, 2013, an invention is deemed to be "described in a printed publication" when it is printed in nearly any kind of document by any means (including electronic means) and has been made available to the public in such a manner that a person of ordinary skill in the art could make the invention. Such availability can occur by circulating copies of a document at a conference, releasing advertising brochures, or disseminating a thesis. Internal business documents or documents provided to others under conditions of confidentiality are not publications within the meaning of the statute. Thus, inventors should place confidentiality notices

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on documents to prevent triggering "publication" of the invention.

When an Invention Is "In Public Use."

Both before and after March 16, 2013, an invention is deemed to be "in public use" if it is being used in the manner intended by the inventor without any confidentiality restrictions. An exception allows **experimental use** of the invention so that the inventor can perfect the invention or ascertain whether it will fulfill its intended purpose. Thus, such experimental use will not defeat novelty. Similarly, private use by the inventor or use for the inventor's own enjoyment will not defeat novelty; however, an invention that is "ready for patenting" will defeat novelty if there is a commercial offer for sale of the invention. *Pfaff v. Wells Elecs., Inc.*, 525 U.S. 55 (1998). In *Pfaff*, the one-year on sale bar of current 35 U.S.C. § 102(b) applied although the inventor had made only engineering drawings and had not yet made the invention, because he had accepted an offer to sell the invention.

An invention is deemed to be "on sale" if it is offered for sale, even though no actual sales occur. Even a single sale or offer to sell may bar patentability.

Nonobviousness

Merely because an invention is useful and novel does not automatically entitle it to patent protection. To qualify for a grant of patent, the invention must be nonobvious to those having ordinary skill in the field or art to which the subject matter pertains. 35 U.S.C. § 103. The subject matter sought to be patented must be sufficiently different from what has been used or described before that it may be said to be nonobvious to a person having ordinary skill in the area of technology related to the invention. For example, the substitution of one material for another in an invention and mere changes in size are ordinarily not patentable because they are obvious. A distinct improvement, however, is patentable even if the new invention improves matter in the public domain. Similarly, a new use of a known process is patentable.

Determining whether an invention is nonobvious is one of the most difficult tasks in patent law. After all, a disposable razor, a safety pin, and a retractable tape measure all seem obvious now, yet none of these items were obvious at the time they were invented.

Until the 1952 Patent Act, courts generally required that an invention result from a "flash of genius" or some sudden insight (an inventor's "aha!" moment). The view was that an invention must have been so nonobvious that no amount of diligent research would have produced it. The Act now provides (and will continue to provide after the AIA) that patentability shall not be negated by the manner in which the invention was made. 35 U.S.C. § 103. Thus, whether an invention is produced by dint of arduous research or a flash of genius does not determine whether it is nonobvious.

The present method of determining **nonobviousness** is by reference to the prior art. **Prior art** is generally defined as all information available to the public in any form about an invention. Until March 16, 2013, when the United States moves to a "first to file" system, prior art is measured from the date of *invention*. On and after March 16, 2013, prior art will be measured from the date of *filing* the application, meaning that information that publicly exists prior to the filing date (other than disclosures by the inventor within one year before filing) will bar the application.

The *Graham* Factors.

In *Graham v. John Deere Co.*, 383 U.S. 1 (1966), the Supreme Court articulated the following factors to consider in determining whether an invention is nonobvious and thus deserving of a patent.

Analogous Prior Art.

One must review the scope and content of the prior art in the pertinent field to determine if an invention is nonobvious. Remember that the term prior art refers to the

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generally available public knowledge relating to the invention for which a patent is sought and that was available prior to invention (or filing, after March 16, 2013). Thus, information contained in existing patents, printed publications, and inventions that were known and used will be considered.

Prior art is pertinent or analogous if it is from the same field of endeavor or from a different field of endeavor but reasonably related to the same problem as that addressed by the invention. *Graham* involved a type of shock-absorber system for a plow shank. Thus, a review of the analogous prior art should consider other plow shanks as well as other shock-absorbing devices (regardless of the field, e.g., shock absorbers used for plow shanks, airplanes, or cars).

Differences between the Prior Art and the Invention.

In determining whether an invention is nonobvious, consideration must be given to differences between the prior art and the invention at issue. Applicants themselves may include statements in their applications in regard to how their inventions differ from and are improvements over prior art. An invention that achieves superior results is likely not obvious.

Level of Ordinary Skill in the Prior Art.

Prior to March 16, 2013, if at the time the *invention was made*, the invention would be obvious to a person having ordinary skill in the art to which the invention pertains, it cannot be patented. After March 16, 2013, if at the time the patent *application is filed*, the invention would be obvious to a person having ordinary skill in the art to which the invention pertains, it cannot be patented. A person of "ordinary skill" is neither a highly sophisticated expert or genius in the art nor a layperson with no knowledge of the field of art, but rather some hypothetical person who is aware of the pertinent prior art.

Secondary Considerations.

In *Graham*, after enumerating the factors evaluated in determining whether an invention is nonobvious, the Court suggested that some **secondary considerations**, all of which are nontechnical and objective, might be considered in determining nonobviousness. Some of the secondary considerations include the following:

- ***Commercial success.*** An invention that is a commercial success may be nonobvious because acceptance by the marketplace tends to show that an invention is significant; moreover, if the invention were obvious, someone would already have attempted to commercialize it for his or her own financial gain.
- ***Long-felt need and failure of others.*** If there has been a longstanding need for a device or process that has gone unresolved despite the efforts of others to solve the problem, and the invention satisfies this need, it tends to show the invention is nonobvious. If the invention were obvious, others would have been able to discover it readily. Similarly, skepticism and incredulity of experts tends to show nonobviousness.
- ***Commercial acquiescence.*** If competitors seek to enter into licenses with the owner so they may use or sell the invention to others, such tends to show nonobviousness because otherwise the third parties would have challenged the patent as invalid based on obviousness.
- ***Copying.*** Copying or infringement of the patent by another suggests nonobviousness because otherwise the infringer would have been able to independently develop the invention. Conversely, the independent and near-simultaneous invention of like products or processes tends to suggest obviousness. If it is easy for a number of people in the field to invent the same product or process, it is likely obvious to those with ordinary skill in the art.

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To ensure that courts do not overrely on the secondary considerations (commercial success, long-felt need, commercial acquiescence, and copying), the Federal Circuit requires that a party demonstrate some link, or *nexus*, between the secondary characteristic and the invention. For example, if an inventor argues that an invention is nonobvious because it is commercially successful, the inventor must demonstrate that commercial success is due to some property inherent in the invention, rather than due to some external factor such as aggressive marketing or the renown of the inventor.

Combination Patents. One area that has caused debate over obviousness relates to new inventions consisting of a combination of older, known elements. The unique combination of the elements may make the invention novel, but the combination must be nonobvious to receive patent protection. A factor sometimes used in determining whether these patents, often called **combination patents**, are nonobvious is *synergism:* The newly combined elements must result in some different function or some result that is different or unexpected such that one who possesses ordinary skill in the art would not have predicted the result of the combination of the known elements. If there is such a new result, the invention is nonobvious and is patentable.

The Patent Act was amended in 1996 to provide preference to inventions relating to biotechnology processes. Section 103(b) expressly provides that a biotechnological process is nonobvious if it uses or results in a novel, nonobvious product or composition. Thus, the process as well as the composition is patentable in the biotechnology field.

There is overlap between the requirements of novelty and nonobviousness. Both depend on examination of prior art. To ensure novelty, one must examine the prior art (what others know, use, and publish) to determine if the invention could have been anticipated. To ensure nonobviousness, one must examine the scope and content of pertinent prior art to determine if the invention would be obvious to a person having ordinary skill in the field.

In 2004, Congress passed the CREATE Act to promote cooperative research and development among universities, the government, and private companies. Under this law (which will be included in AIA's revised Section 102), information disclosed by one entity to another (for example, by a university employee to a private company) will not create a bar to patentability as prior art. Their work is deemed owned by the same entity and not subject to an obviousness objection as prior art if certain conditions are met (such as having a joint research agreement).

In one of the most important patent rulings in a generation, in 2007, the U.S. Supreme Court tightened the obviousness standards for obtaining patents on new products that combine elements of already existing inventions. In *KSR International Inc. v. Teleflex Inc.*, 550 U.S. 398 (2007), a unanimous Court held that if the combination results from nothing more than "ordinary innovation" and does nothing more than yield "predictable results," the invention is obvious and is not entitled to patent protection. The Court held that the Federal Circuit's test (which had held that if there was a "teaching, suggestion, or motivation" in the prior art that would have led a person of ordinary skill in the relevant field to combine known elements, the resulting invention should be barred for obviousness) had been too strictly applied and often required third-party documentation. The Court endorsed a more "common sense" and flexible approach: If a person of ordinary skill in the art can implement a predictable variation of an invention and would see the benefit of doing so, then the invention should be barred for obviousness. While the "teaching, suggestion, motivation" test provides helpful insights (and was not eliminated by the ruling in *KSR*), it cannot be used in a rigid and inflexible manner. In determining obviousness,

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ETHICS EDGE

COMPETENT LEGAL RESEARCH

The duty to perform adequate legal research has been discussed in several cases. Legal professionals are expected to know common principles of law and to be able to discover other rules of law that may readily be found by standard research techniques. *Camarillo v. Vaage*, 130 Cal. Rptr. 2d 26, 32 (Ct. App. 2003). Thus, patent paralegals must be familiar with basic patent principles relating to novelty and nonobviousness.

one must thus ask whether the improvement is more than a predictable use of prior art elements according to their established functions. Accordingly, if at the time of invention there is a known problem for which there is an obvious solution, an invention should be barred for obviousness. Conversely, if the elements of an invention work together in an unexpected and fruitful manner, then the invention is likely not obvious. Most experts correctly predicted that the ruling in *KSR* would have far-reaching implications and would make obtaining patents more difficult (because patent examiners will reject applications for patents on the basis that the invention results from nothing more than ordinary innovation and is thus obvious). In fact, the USPTO has issued guidelines for its examiners to help them determine obviousness in the wake of *KSR*. Paradoxically, although this tightening of the obviousness test may stem the tide of junk patents, it may also produce litigation challenging existing patents on the basis that they are obvious. In fact, in just the first four years after the decision in *KSR*, more than 600 court cases and thousands of administrative decisions had cited it.

Exclusions from Patent Protection

There are a variety of items, products, and processes that cannot be patented. The following are excluded from patent protection either by statutory prohibition or judicial interpretation:

- ***Products of nature.*** Only human-made inventions can be patented. Naturally occurring substances cannot be protected by patent even if they have previously been unknown to others. For example, a new plant or mineral that is discovered cannot be patented. On the other hand, genetically altered living organisms can be patented as "manufactures" or "compositions of matter." In *Diamond v. Chakrabarty*, 447 U.S. 303 (1980), the Court held that a live, human-made microorganism (in that case, a genetically engineered bacterium capable of breaking down crude oil) was patentable as a human-made "manufacture" or "composition of matter" and stated that "anything under the sun made by man" is patentable. Thus, genetically altered oysters and mice have been held patentable. The distinction is not between living and inanimate things but between products of nature (whether living or not) and human-made inventions. *Id.* at 313. The new AIA expressly prohibits the patenting of human organisms.
- ***Laws of nature.*** Laws of nature, physical phenomena, scientific truths, and abstract ideas cannot be patented. As the Court remarked in *Chakrabarty,* "Einstein could not patent his

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celebrated law that $E = mc^2$; nor could Newton have patented the law of gravity." *Id*. at 310. Generally, mere chemical or mathematical formulas or algorithms divorced from any tangible result cannot be patented; however, chemical substances can be patented, systems using formulas can be patented as processes, and items that are produced as the result of formulas can be patented. Thus, a machine or product that depends on some law of physics or the law of gravity can be patented. One example given by the USPTO is that the concept of magnetism is not patentable, but a magnetic door latch (the practical application of the concept) is patentable. A process is not unpatentable merely because it contains a law of nature or a pure mathematical algorithm. Similarly, computer programs, although they consist of algorithms, can be patented in many instances. In *Diamond v. Diehr*, 450 U.S. 175 (1981), a patent was upheld for the process of molding rubber products with the aid of a computer. The Court noted that although mathematical formulas as such are not patentable, a process employing a well-known mathematical equation was patentable. Additionally, the use of a computer to improve an otherwise patentable process did not preclude the process from being patented. Thus, incorporating a computer program as a step in a process or as a component in a machine or manufacture may be acceptable. In sum, a disembodied mathematical concept that represents a law of nature or abstract idea is not patentable; however, if the principle is either tied to a particular machine or apparatus or transforms a particular article into a different state or thing, it *may* be patentable. See Chapter 21 for additional discussion on protection of computer programs under patent law.

- ***Printed matter.*** Printed forms cannot be patented.
- ***Atomic weapons.*** Under the Atomic Energy Act of 1954, atomic weapons cannot be patented.
- ***Business methods and mental steps.*** Systems for the operation of businesses cannot be patented unless they are more than mere abstract ideas. For example, in a key patent decision, in 2010 the U.S. Supreme Court held that a method by which energy consumers could protect themselves against the risk of price fluctuations (namely, a method of hedging risk) was not patentable because it was a mere abstract idea. *Bilski v. Kappos*, 130 S. Ct. 3218 (2010). Similarly, processes consisting solely of mental steps, meaning human thought and deliberation, cannot be patented. Finally, under the AIA and effective September 16, 2011, any strategy for avoiding, reducing, or deferring tax liability cannot be patented because it is deemed prior art. See Chapter 21 for additional discussion on this topic.

DESIGN PATENTS

The Patent Act provides that whoever invents any new, original, and ornamental design for an article of manufacture may obtain a patent. 35 U.S.C. § 171. Comparing the scope of this provision to that of Section 101 relating to utility patents, it is clear that the requirement of "usefulness" for utility patents has been replaced by a requirement of "ornamentality." The design must be novel and nonobvious. Just as a person may obtain a patent for a chemical compound and one for the process utilizing the chemical compound, a person may obtain a utility patent and a design patent for one article of manufacture because a useful article may be ornamented with a design. A **design patent** can be obtained for articles as diverse as jewelry, furniture, trash receptacles, and clothing (such as an ornamental design for a shoe).

Recognizing that while authors could seek protection for their works under copyright law and inventors could seek protection for their works

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under patent law, no protection was afforded for the creators of decorative arts, Congress filled this gap and allowed design patents in 1842. One of the first design patent cases related to a claim by Gorham Manufacturing that its spoon and fork handle design patents had been infringed. *Gorham Mfg. v. White*, 81 U.S. (14 Wall.) 511 (1871). In *Gorham*, the Court noted that the intent of the design patent statute was to encourage the decorative arts.

To be protectable a design must satisfy the following four requirements:

1. ***It must be an article of manufacture.*** An **article of manufacture** is nearly anything made by a human and may consist of a manufactured article's configuration (the particular shape of a chair), surface ornamentation (the design on the handle of tableware), or a combination of both (a uniquely shaped fork with two irregularly sized tines, embossed with a leaf design).
2. ***It must be new.*** The requirements of novelty for utility patents discussed in the previous section on patentability under the heading "Novelty" apply equally to design patents.
3. ***It must be original.*** The requirement of originality is often viewed as the equivalent of the requirement of nonobviousness for utility patents. A design cannot be patented if a designer of ordinary skill who designs articles similar to the one applied for would consider the design obvious in view of the prior art.
4. ***It must be ornamental.*** To be patentable, a design must be primarily ornamental rather than utilitarian. The article may serve a useful purpose (e.g., a spoon and a chair serve a useful purpose), but its primary purpose cannot be functional. A design or shape that is entirely functional, without ornamental or decorative aspect, does not meet the criteria for a design patent. If there are several ways that an item could be designed and yet still remain functional, then any one design is likely ornamental rather than functional. An article may qualify for design patent protection if at some point in its life cycle its appearance is a factor even if, when ultimately used, the article is hidden. Thus, caskets and artificial hip prostheses have been given design patent protection. Although both are primarily useful, the appearance of the article is of importance at some point, even though when put to their intended use, the articles are hidden from view. The bars to novelty set forth in Section 102 and discussed earlier in this chapter apply equally to design patents whether before or after the relevant provisions of the AIA take effect. For example, after March 16, 2013, a design patent will be awarded to the first inventor to file the application for it (and the inventor has a one-year grace period after his or her disclosure of the invention to file the application). Most of the bars aim at ensuring that designs are promptly patented.

There is some overlap between design patents and copyright. It has been held that the same work can qualify for protection under both copyright law and patent law. For example, the drawing of a cartoon character can be protected under copyright law; when the character is embossed on some article, such as china plates, it may be protected under patent law. Similarly, the same item may qualify for protection under trademark law and patent law. For example, in a case involving the shape of a bottle of Mogen David wine, the court stated the design qualified for both patent and trademark protection. *In re Mogen David Wine Corp.*, 328 F.2d 925 (C.C.P.A. 1964). Finally, there is overlap between trade dress protection and patent law such that the ornamental appearance of an item may be protected under trade dress law and as a design patent. Securing copyright protection is generally the easiest and least expensive method of protecting

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certain designs. Although copyright registration is not required, a copyright registration may be obtained for a filing fee of $35 (for online filing) within several months, and protection will last for the life of the author plus 70 years. Trademark registrations can last forever (if they are properly maintained and renewed), and patent protection comes into being only once the patent issues, a process that may take three years or longer from the date of application. Moreover, the nonrenewable term of a design patent is 14 years from the date of issuance of grant. Practical matters should also be considered. If a design will likely be somewhat limited in its appeal or trendy, such as a piece of jewelry, it is far less expensive and far more expeditious to obtain copyright rather than patent protection for the item. Thus, a combination of intellectual property strategies may be needed to secure the broadest possible scope of protection for certain articles.

PLANT PATENTS

Patents for plants have been recognized only since the passage of the Plant Patent Act in 1930. Prior to that time, the philosophy was that plants were natural products not subject to patent protection. Section 161 of the Patent Act now provides that whoever invents or discovers and asexually reproduces any distinct and new variety of plant may obtain a patent therefor. Just as design patents substitute the requirement of "ornamentality" for the "usefulness" required of utility patents, plant patents substitute the requirement of "distinctiveness" in place of "usefulness."

Congress allowed patents for plants to provide the benefits of the patent system that were then available to manufacturing and industry to the agriculture business in order to incentivize growers and protect their plant products from infringement. Without patent protection, copies of plants could be produced by grafting techniques. Thus, a **plant patent** affords its owner the right to exclude others from asexually reproducing the plant. The term of protection is the same as that for utility patents, namely, 20 years from the patent application date. The term *plant* is construed in its ordinary meaning and does not include bacteria. The use of the term *discovery* means a discovery by asexual propagation; plants discovered in the wild cannot be patented; however, a plant discovered in a cultivated area and later reproduced asexually can be patented. Tubers, such as Irish potatoes, are not eligible for protection under the Plant Patent Act (due to their importance as a food source).

There are four requirements that must be satisfied before a plant can be patented:

1. ***The new variety must be asexually reproduced.*** A grower or discoverer of a new and distinct variety of plant must be able to reproduce the plant by asexual means. Generally, this is accomplished by taking cuttings of the original plant and placing them in soil or by grafting so as to create a new plant. **Asexual reproduction** involves growing something other than from a seed.
2. ***The plant must be distinctive.*** The new plant must be clearly distinguishable from existing varieties. Features that show distinctiveness are color, odor, flavor, shape, ability of the plant to grow in a different type of soil, its productivity, its factors of preservation, or immunity to disease. The requirement of distinctiveness imposes only a requirement that the plant be different from, rather than superior to, other varieties. A new color of a rose may be different from other roses but may not be better than some other rose variety. Nevertheless, it may be patented. There is a great deal of overlap between nonobviousness for utility patents and distinctiveness for plant patents. One patent

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for a variety of bluegrass recites that the plant is a "new and distinct variety of Kentucky Bluegrass characterized by its excellent tolerance to drought, low fertilizer requirements, deep rooting system, excellent tolerance to Fusarium blight, and good to excellent shade tolerance. The plant tolerates a close cut, is highly resistant to most common bluegrass diseases, is extremely aggressive, has a medium to coarse leaf texture and consistently maintains excellent turf quality." U.S. Plant Patent No. PP4,704.

3. ***The plant must be novel.*** The new variety of plant must not have previously existed in nature. The bars to novelty set forth in Section 102 and discussed in the earlier section on patentability under the heading "Novelty" apply equally to plant patents (whether before or after the enactment of the AIA). For example, after March 16, 2013, a plant patent will be awarded to the first inventor to file the application for it (and the inventor has a one-year grace period after his or her disclosure of the invention to file the application).
4. ***The plant must be nonobvious.*** Although Section 161 of the Patent Act providing for statutory protection for plant patents does not specifically state that patents for plants must be nonobvious, the section does provide that the provisions of the Act relating to utility patents apply equally to plant patents unless otherwise specified. Generally, the standards set forth in *Graham v. John Deere Co.*, discussed in the earlier section on patentability under the heading "Nonobviousness," apply equally with regard to plant patents.

See Exhibit 17–2 for comparison of utility, design, and plant patents.

Although the most common means of securing protection for plants is through application for a plant patent, two other methods of protection exist for plants. In 1970, Congress enacted the **Plant Variety Protection Act** (7 U.S.C. § 2321),

	Utility Patent 35 U.S.C. § 101	**Design Patent 35 U.S.C. § 171**	**Plant Patent 35 U.S.C. § 161**
Inventions Covered	Processes, machines, manufactures, compositions of matter, or improvements thereof	Designs for articles of manufacture	Plant varieties
Usefulness Required	Yes	No	No
Novelty Required	Yes	Yes	Yes
Nonobviousness Required	Yes	Yes (referred to as "originality")	Yes
Ornamentality Required	NA	Yes	No
Asexual Reproduction Required	NA	NA	Yes
Distinctiveness Required	NA	NA	Yes

EXHIBIT 17–2 Comparison of Utility, Design, and Plant Patents

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allowing quasipatent protection for certain sexually reproduced plants, meaning plants bred through seeds. Sexually reproduced plants could not receive protection under the 1930 Plant Patent Act because new varieties could not be reproduced true-to-type through seedlings. The new variety was not always stable, and its new characteristics could not be passed uniformly from one generation to the next. By 1970, however, it was generally recognized that true-to-type sexual reproduction of plants is possible, and the Plant Variety Protection Act was passed.

These new varieties, produced through the use of seeds, are awarded "plant variety protection certificates" by the Department of Agriculture, rather than letters patent issued by the USPTO. The primary purpose of the Plant Variety Protection Act is to encourage the development of novel varieties of sexually reproduced plants. The certificate owner may exclude others from selling, offering, reproducing, or trading infringing plants. The term of protection is generally 20 years from date of issue of the certificate.

Finally, a plant may qualify for protection as a utility patent, even though it also qualifies for protection under the Plant Patent Act as a plant patent or under the Plant Variety Protection Act for patent-like protection. An application for a utility patent for a plant must satisfy the requirements for utility patents generally, namely, usefulness, novelty, and nonobviousness.

In some cases, matters of tactics and strategy may dictate whether an inventor seeks protection for a plant under the Plant Patent Act of 1930, the Plant Variety Protection Act of 1970, or as a general utility patent. Although the most significant determinant is how the plant can be reproduced (asexual reproduction is required under the Plant Patent Act while sexual plant reproduction is protected under the Plant Variety Protection Act), other factors should be considered. For example, the requirement that an invention be nonobvious applies to utility patents and plant patents reproduced asexually, but not to the Plant Variety Protection Act. Thus, an inventor who cannot demonstrate nonobviousness might seek protection under the Plant Variety Protection Act. Similarly, fees to maintain the patent are due on utility patents but not on plant patents. Assuming applications do not claim identical subject matter, it is possible that a patent could be obtained for a plant both under the Plant Patent Act and under the Patent Act as a utility patent. Each statute has different requirements and affords different protections.

DOUBLE PATENTING

The **double patenting** principle prohibits the issuance of more than one patent for the same invention or for an invention that is substantially the same as that owned by an inventor. The intent of this bar is to ensure that inventors do not make some insignificant change to an invention near the end of its term of existence in order to secure another 20-year monopoly on the invention or discovery. Nevertheless, design and utility patents may coexist for the same item because they relate to arguably distinct subject matter. A utility patent claims protection for the usefulness of the object while a design patent covers its ornamentality. If the USPTO refuses a patent application on the basis of double patenting, inventors often respond by inserting a **terminal disclaimer** into their applications, agreeing that the term of protection for a second invention will terminate upon expiration of the patent for the first invention and that the patent claim to the second invention will be valid only as long as both inventions are owned by the same person or entity. Both patents will thus expire on the same day. Such a disclaimer will enable an inventor to overcome a rejection based on double patenting. Double patenting is discussed further in Chapter 18.

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THE ORPHAN DRUG ACT

Just as the Plant Variety Protection Act grants rights somewhat similar to patents, the Orphan Drug Act (21 U.S.C. §§ 360aa–360ee) provides rights for certain drugs that are similar to patent rights, namely, an exclusive right for seven years to market an **orphan drug** that is necessary to treat a disease that affects fewer than 200,000 people. Protection under the Act is triggered when the Food and Drug Administration determines that unless such protection is granted, the drug would likely not be made or available to those in need of it. The Act applies whether or not the drug can be patented (thus affording protection for drugs that may lack novelty or nonobviousness). The Act thus provides incentives to pharmaceutical companies to develop drugs for rare diseases or conditions when otherwise they might not invest the time and effort in developing a drug for a small target group if patentability could not be assured.

TRIVIATRIVIATRIVIATRIVIATRIVIATRIVIATRIVIATRIVIATRIVIATRIVIATRIVIATRIVIATRIVIA

- Most experts attribute the modern concepts of patents to England, where, in 1449, King Henry VI granted a patent for manufacturing stained glass to John of Utynam.
- Between fiscal years 2004 and 2011, the number of patent applications filed increased 15 percent.
- More than 94 percent of the patent applications filed with the USPTO are for utility patents; approximately 5 percent are for design patents; and less than 1 percent are for plant patents.
- George Washington Carver invented more than 300 uses for peanuts.
- The inventor of VELCRO® reported that he thought of the invention while removing burrs from his pet's fur after walking in the woods.

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CHAPTER SUMMARY

A patent is a grant from the federal government allowing an inventor to exclude others from producing, using, selling, or importing the inventor's discovery or invention for a limited period of time, generally 20 years from the filing date of a patent application. Patent laws are aimed at fostering and promoting discoveries. There are three types of patents: utility patents, design patents, and plant patents. The vast majority of patents are utility patents, which must satisfy the requirements of usefulness, novelty, and nonobviousness in order to secure protection. An invention or discovery that is merely an insignificant addition to or trivial alteration of something already known or in existence is unpatentable. Protection is allowed for processes, machines, human-made articles of manufacture, and compositions of matter.

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Not all inventions or discoveries may be patented. Additionally, patents are not available for scientific or mathematical principles, some business methods, printed matter, or substances existing in nature. Generally, until March 16, 2013, patents will be awarded to the first to invent. After that date, the United States will move to a "first to file" system, and patents will be awarded to the first to file the application for the claimed invention.

Design patents protect original, new, and ornamental designs for articles of manufacture. Plant patents protect new and distinct varieties of asexually reproduced plants.

CASE ILLUSTRATION

EXPERIMENTAL USE

Case: *TP Laboratories, Inc. v. Professional Positioners, Inc.*, 724 F.2d 965 (Fed. Cir. 1984)

Facts: Plaintiff sued for infringement of its patented orthodontic device, and defendants asserted that the patent was invalid because the invention was in public use more than one year prior to the patent application therefor. The district court held that the invention was in public use and was thus invalid. Plaintiff appealed, alleging that its use was experimental use.

Holding: Reversed. The invention, a molded tooth positioning appliance, was used on at least three of plaintiff's patients more than one year before the patent application for the device was filed. However, the Federal Circuit held that such use was experimental. It was not a public use because disclosure of the device to the patients who used or tested it could not be avoided. Moreover, there were no sales or commercial exploitation of the device at any time before the patent application was filed. Such showed that the device was merely being tested rather than commercially used.

CASE STUDY AND ACTIVITIES

Case Study: Holiday's engineers have invented a new device that more accurately tracks passengers as they embark and disembark its ships. The device was invented on April 14, 2011, although a patent application was not filed for the device until May 20, 2012, because Holiday was testing the device on three of its ships to ensure it worked properly. Holiday began offering the device for sale only after it filed its patent application. The patent was granted. Also, Holiday's design team recently developed a new design for a massage mat to be used in the ship spas.

Activities: Discuss whether the patent for the tracking device is valid and whether the massage mat is patentable.

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ROLE OF PARALEGAL

Until patent searching begins and an application is filed, the role of an IP paralegal will be somewhat limited. Typically, tasks may consist of conducting research to help assess and satisfy the required elements of usefulness, novelty, and nonobviousness. If a law firm does not have a general information letter available that can be sent to clients who have basic questions about patent law, one should be drafted. Similarly, a "frequently asked questions" sheet can be prepared for distribution to clients. Flowcharts showing the patent process can be made, and charts comparing types of patents (e.g., plant patents and utility patents for plants) can be prepared.

An IP paralegal may also be involved in conducting some type of audit of clients to inquire whether they have invented any processes or products that may qualify for patent protection. A questionnaire should be prepared to assess whether clients' intellectual property is being fully protected. See Chapter 24 for sample questions for an intellectual property audit.

INTERNET RESOURCES

Federal laws relating to patents:	http://www.law.cornell.edu http://www.gpo.gov/fdsys
U.S. Patent and Trademark Office:	http://www.uspto.gov (general information about patents [including the provisions of the new AIA], frequently asked questions, patent searching, forms, and filing fees)
Manual of Patent Examining Procedure:	http://www.uspto.gov/web/offices/pac/mpep/mpep.htm (Chapter 700 contains information on rejections based on lack of novelty, lack of usefulness, lack of nonobviousness, and double patenting)
National Inventors Hall of Fame:	http://www.invent.org (honoring inventors and their inventions)
General information:	http://ipmall.info (Franklin Pierce IP Mall with excellent information and articles and links to other IP sources) http://www.megalaw.com http://www.findlaw.com
Glossary of IP terms:	http://www.uspto.gov/main/glossary
Absurd patents:	http://www.totallyabsurd.com (identifies odd and unusual patents)

DISCUSSION QUESTIONS

1. Classify each of the following as a utility, design, or plant patent.
- a remote control for a television
- string for a tennis racket
- a picture frame with lattice trim

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- an early maturing apple tree
- jewelry with attachable charms
- a kitchen timer
- headgear or helmet for football

2. Ben sees a new plant while walking in the mountains. Is the plant patentable? Discuss.

3. A corporation has released a memo to its 10 most senior vice-presidents, describing the corporation's new invention. Will this publication bar an application for a patent filed more than one year later on the basis that the invention was described in a printed publication under 35 U.S.C. § 102? Discuss.

4. Sarah invents a type of cartridge for ink for copy machines. The cartridge is identical to previous cartridges invented by others, although it is slightly smaller. Is the invention patentable? Discuss.

5. Effective March 16, 2013, the United States will move from its current system of granting patents to the "first to invent" the invention to the "first to file" the application for the invention (with a grace period for the inventor's own disclosures). Discuss the advantages and disadvantages of this change.

6. Discuss whether the following items are likely patentable or what objections the USPTO might raise to a patent application for these items:

- a printed form for keeping baseball scores
- a surgical scalpel
- an idea for defeating identity theft
- a computer program for defeating identity theft
- a pharmaceutical composition for treating lung cancer
- a transgenic pig with altered hormone receptors
- a pig

7. On January 1, 2009, Samantha filed a patent application for a utility patent, a design patent, and a plant patent. All patents were granted on January 1, 2012. When will each patent expire?

USING INTERNET RESOURCES

1. Access Section 716.05 of the Manual of Patent Examining Procedure, available through the website of the USPTO. What secondary consideration might show strong evidence of nonobviousness?

2. Access the Glossary available through the website of the USPTO. What is a "terminal disclaimer"?

3. Access the website for the USPTO, locate the following patents, and give a brief description of each: Patent Nos. 7,325,329, D638,633, and PP9,013.

Go to http://www.paralegal.delmar.cengage.com for Quizzes, Forms, Chapter Resources, and additional information.

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CHAPTER 18

Patent Searches, Applications, and Post-Issuance Proceedings

CHAPTER OVERVIEW

Before an application for a patent is filed, a search should be conducted to ensure that the invention is novel and nonobvious. If the search results suggest that an invention may be patentable, an application is then prepared. An application consists of two parts: the specification (describing the invention) and the inventor's oath or declaration. Until September 16, 2012, applications must be filed by individual inventors, although the application can be assigned to another at the same time it is filed. After the application is filed at the USPTO, it will be examined for patentability. Application proceedings at the USPTO are confidential until the application is published, generally 18 months after the application filing date. The examiner may issue office actions, requiring amendment of some of the claims of the invention. No new matter can be added to an application. When an application is allowed by an examiner, a notice of allowance is issued, and an "issue fee" must be paid to the USPTO for the patent to be granted. The term of utility and plant patents is 20 years from the date of filing of the application therefor. The term of design patents is 14 years from the date of grant. Maintenance fees must be paid at three intervals during the term of a utility patent to maintain it in force. Once the patent is issued, its owner may exclude others from making, selling, importing, or using the invention for the term of the patent.

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PATENT SEARCHING

The Need for a Search

Patentability requires novelty and nonobviousness. The only predictable method of determining whether an invention is new and nonobvious is to conduct a search of the prior art (including patent records and printed publications). The patentability search, sometimes called a **novelty search** or prior art search, will help determine whether the differences in the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious to a person having ordinary skill in the art. Moreover, because 35 U.S.C. § 102 imposes novelty as a condition for patentability, a search will disclose whether novelty bars to protection exist. Finally, if an invention has fallen into the public domain because its patent has expired, anyone can use it and no one can obtain a patent for it. Searching will disclose the existence of such expired patents. Thus, although not required prior to filing a patent application, a search is recommended to determine the feasibility of obtaining a patent. Otherwise, an inventor may incur costs of several thousand dollars in prosecuting a patent application only to have an examiner determine that the invention fails to satisfy the requirements of novelty and nonobviousness. An additional benefit of conducting a search is that it may provide ideas for drafting the application itself.

A novelty search is somewhat limited in scope and is designed to disclose whether an application will be rejected on the basis of lack of novelty or obviousness. A basic novelty search can usually be completed for less than $2,000, although pricing varies depending on the type and complexity of the invention. If an invention is intended for immediate commercial use or sale, an additional search, called a freedom to operate or infringement search, is often conducted concurrently with the novelty search. An invention may be patentable as a significant improvement over an existing invention and yet still infringe a patent.

Although the terms "freedom to operate search" and "infringement search" are often used interchangeably, there are differences between the two types of searches, although both focus on whether the client's invention is blocked by another's intellectual property rights. A **freedom to operate** search is usually conducted before the invention is brought to market and is intended to ensure that the invention does not infringe any patents. In contrast, an **infringement search** is usually conducted after the inventor has been informed that he or she is violating another's patent, and it focuses on examination of this specific and known patent (often with the goal of invalidating this patent).

An infringement search or full patentability search is far more extensive than a novelty search and is thus more expensive, often costing between $3,000 and $10,000. Conducting a search and obtaining an opinion relating to infringement is important because although there is no affirmative duty to obtain advice of counsel with regard to whether one's actions might constitute infringement, relying on counsel's advice is an important factor in determining willfulness. Generally, willful infringement may lead to the imposition of punitive damages in an infringement action (see Chapter 20).

Some inventors conduct their own searches. Others retain patent attorneys to perform the search or to engage the services of a professional search company. Naturally, the scope and breadth of the search depends on a variety of factors, including cost, complexity, and importance of the invention. In many instances, paralegals conduct preliminary searches of the U.S. Patent Trademark Office (USPTO) databases and if this initial review indicates that the invention may be patentable, a more comprehensive search is conducted by professional patent searchers.

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Patent Search Resources

There are several separate resources or facilities for patent searching.

USPTO Public Search Facility.

The USPTO maintains a Public Search Facility at its Alexandria, Virginia offices. The search room is open to the public and houses all U.S. patents granted since 1790. Searching is usually accomplished by using state-of-the-art computer databases (described later). Trained staff is available to assist searchers. The USPTO also offers a Scientific and Technical Information Center at its offices, which offers more than 120,000 volumes of scientific and technical books as well as numerous journals and foreign patents (although most of the collection now consists of electronic materials).

Patent and Trademark Depository Libraries.

Because many inventors do not have the resources to travel to the USPTO's Virginia offices, the USPTO has designated more than 80 libraries throughout the nation as Patent and Trademark Depository Libraries (PTDLs). Nearly every state has a PTDL, and larger states, such as California and New York, have several. About one-half of the PTDLs are academic libraries affiliated with universities, and the other one-half are public libraries. The PTDLs receive copies of patents and offer free public Internet access to all USPTO search tools, indices, and directories. The scope of the print collections varies from library to library. A list of all PTDLs is available on the USPTO website.

USPTO Online Database.

The USPTO's online database includes all patents issued since 1790 and applications published since January 2001. The database of more than eight million patents includes information about all U.S. patents, offering the full text of patents granted since 1976, and the patent number and current U.S. classification for all patents granted from 1790 through 1975. Copies of patents may easily be ordered through the USPTO website or via telephone upon payment of a moderate fee.

USPTO Search Assistance.

The USPTO public search facility staff conducts minimal searching for a fee. Specifically, the staff will conduct an inventor search and then provide a list of the patent numbers related to that inventor for $40 per hour.

Commercial Search Services.

Although many paralegals and patent agents and attorneys (and some inventors) are skillful in conducting patent searches using the USPTO databases, most patent attorneys use or recommend the use of a professional commercial patent search company for the most accurate and complete patentability and infringement searches. In many instances, a combination approach is used: The paralegal or patent attorney performs a preliminary search using online databases, and if the results suggest that the invention is patentable, a more comprehensive search is then ordered from a commercial vendor. These vendors have numerous databases at their fingertips and are highly experienced in locating the relevant prior art. Some search companies simply provide access to their vast databases for a fee charged to the inventor or law firm; others perform the search themselves and provide a full written report. Searches can be customized to the inventor's needs; for example, a search may focus solely on foreign patents. Some of the well-known commercial search companies include the following:

Delphion. Originally a product of IBM and now owned by Thomson Reuters (the provider of Westlaw), Delphion (http://www.delphion.com) provides a full complement of patent searching options. It is highly popular and allows searching by patent

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number, inventor name, and a variety of other fields. It offers monthly subscriptions and a "one-day pass" for private inventors.

Dialog. Dialog provides access to hundreds of databases and more than 15 million patents covering 60 countries. Dialog (http://www.dialog.com) offers access to IP firms and practitioners who then perform searches to determine patentability.

Westlaw. Subscribers to Westlaw (http://www.westlaw.com) may access its comprehensive database Westlaw Patents for "one- stop shopping" for all information relating to patents, including patent file histories and graphical views of both "patent family trees" to see connections between multiple patents and patent claims to view the evolution of claims in a patent.

LexisNexis. LexisNexis (http://www.lexisnexis.com/patentservices) offers a Patent and Trademark Solutions service. You may obtain copies of U.S. and foreign patent documents, file histories from the USPTO, and other documents. Moreover, professional searchers at LexisNexis perform prior art searches and produce a written search report.

MicroPatent. MicroPatent (http://www.micropat.com) is yet another Thomson Reuters business. Its PatentWeb provides inventors and law firms electronic access to tens of millions of global patent documents. Searching can be done by keyword, patent number, and various other means.

PatPro, Inc. PatPro (http://epatpro.com) is located near the offices of the USPTO. It was founded by former USPTO examiners and it offers document retrieval and full search reports that list references found and discuss those references.

The previously mentioned commercial vendors are just a few of the many companies that perform patent searches and related services. Links to many professional search companies can be found at the "Patent Searching Academy" offered by Franklin Pierce's IP Mall at http://www.ipmall.org/web_resources/record_request_pb_12.php The IP Mall is one of the best-known IP sites on the Internet. It provides a wealth of IP resources, links to other useful IP sites, articles, information, and many other reliable and useful tools.

In addition to using the free USPTO online patent search databases, those interested in performing patent searches may also use Google's new and free Google Patent Search. All of the patents available through Google come from the USPTO's database, and one may search by keyword (similar to any Google type search) or by criteria such as inventor's name, patent number, and filing date. Access the website at http://www.google.com/patents.

Patent Search Methods

There are two primary methods that can be used to search for prior art: the keyword search method and the classification search method.

Keyword Searching.

The patent record databases (e.g., the USPTO's database of more than eight million issued patents) allow searching by keyword. This method matches words, phrases, and terms relating to the claimed invention to the words, phrases, and terms in the patents themselves. Patents from January 1976 to the present can be searched by a variety of fields or terms, such as the inventor's name, the patent's title, the full description of the invention, and the claims.

Most keyword searches rely on Boolean searching to formulate queries. The **Boolean search** method uses terms and connectors such as *or, and,* and *and not* to construct searches. For example, a query of "mouthpiece and clarinet" would produce documents in a database only if both of those words were present

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in the document. The USPTO's "Quick Search" screens allow searchers to use the Boolean connectors to construct search queries using easy-to-use drop-down menus, so searchers may search patents by inventor name, title of invention, and so forth.

Additionally, the USPTO database affords several elements or "fields" that can be searched, so that a search will retrieve only documents relating to the attorney representing an applicant, a specific patent examiner, issued patent number, and so forth. The display of each patent's full-text includes a hyperlink to obtain full-page images of each page of the patent. Pre-1976 patents can only be searched by the patent number or the USPTO's classification number or code assigned to the invention; however, this limited display also includes a hyperlink to obtain full-page images of each page of the patent.

The USPTO database affords several methods to narrow a search and obtain precise results. For example, searchers may select a date range to obtain patents issued only after a specific date.

Keyword searches are fast and easy; however, the quality of a keyword search is highly dependent on the searcher's ability to anticipate the words an applicant used in an application. For example, a keyword search for a term such as "bird" will produce only patents with that specific word and no patents with the word "avian."

Introduction to Classification Searching.

For more than 100 years, the USPTO has used a classification system, called the **United States Patent Classification**, for its patents. Although the system is primarily designed to be used by patent examiners in the course of examining patent applications, the system is also used by searchers. In fact, most commercial searchers use the classification method of searching. The system categorizes inventions according to the features of the invention. All relevant patents for a given technology are grouped together by class. There are separate classes for design and plant patents.

There are approximately 450 classes of inventions, from class 005 for beds, to class 102 for ammunition and explosives, to class 433 for dentistry. Thus, any inventions relating to beds are found in class 005. Within each class are subclasses, which are smaller, more refined subsets within the main class. Thus, a designation such as 005/110 refers to the main class of beds and the subclass of cots; the designation 005/665 refers to the class of beds and the subclass of waterbeds. The USPTO uses three digits for all classes and three digits for all subclasses. Some inventions may be described by multiple classification numbers or codes. There are approximately 150,000 subclasses. Most classes have about 300 subclasses.

Searchers often begin with the Index to the U.S. Patent Classification, which functions much like the index at the back of this text. It is an alphabetical listing of technical and common terms from abacus (in class 434 for education and demonstration) to zwieback (in class 426 for food and edible material). The Index provides a useful introduction to the classification system.

Classification searches are useful because they retrieve all patents issued since 1790. Moreover, they do not require searchers to "guess" the words used to describe an invention as does the keyword technique. However, the classification system can be daunting for newcomers, who will need to spend some time learning how the system works. Moreover, as new technologies are developed, the USPTO assigns new classes and subclasses, which can make it difficult for new searchers to locate patents in cutting-edge technologies. Additionally, sometimes patents are misclassified. Finally, note that the classification system covers only U.S. patents; foreign patents cannot be obtained through the USPTO database.

Explanations of the U.S. Patent Classification system and suggestions for searching are provided on the USPTO's website at http://www.uspto.gov/web/patents/classification/help.htm.

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Using the Classification Method to Search Patents. The USPTO's website suggests the following four-step strategy to search for patents using the classification system (all steps may be accomplished online).

1. Begin with the alphabetical listing in the Index to the U.S. Patent Classification (located at http://www.uspto.gov/web/patents/classification/uspcindex/indextouspc.htm) to "get your feet wet" and gain a basic understanding of how the system works. Look for common terms describing the invention, its function, and use. Note the class and subclass numbers.
2. Use the U.S. Manual of Classification (located at http://www.uspto.gov/web/patents/classification) to access the classes and subclasses you located in the Index. The Manual provides lists of class titles in both numerical and alphabetical order. The titles are descriptive and suggestive of the technology involved and will further help refine the search parameters. The Manual's list of classes and subclasses is helpful in showing interrelationships between classes and subclasses.
3. Review the Classification Definitions (which are comprehensive descriptions of the class, located at http://www.uspto.gov/web/patents/classification). These definitions establish the scope of the class and subclass and provide important search notes and suggestions for further research.
4. Click on the "P" icon to automatically search and retrieve patents for that particular classification code. Review the patents retrieved to make sure you are on the right track.

Other Search Methods. As described, patents in the USPTO database (and other databases) can be accessed by a variety of fields. Thus, for patents issued since 1976, searchers can locate all applications filed by a certain inventor or all patents owned by a certain individual or company. Searchers can also locate any patent by its patent number. Because many inventions and products are marked with their patent numbers and websites and technical and scientific literature will note patent numbers, locating patents by number can be useful in reviewing similar or related prior art. Similarly, a product may be marked with the notice "patent pending." A searcher can then try to locate information about the invention by searching for its inventor or owner. Most applications are published 18 months after their filing dates; thus, information about these pending applications may be available.

Other websites offer highly specialized patent searching. For example, the U.S. Department of Agriculture offers searching of some patents related solely to the intersection of agriculture and biotechnology at its website at http://www.ers.usda.gov/Data/AgBiotechIP.

Limitations on Patent Searches. Although a thorough search by an experienced searcher can provide invaluable information, there is no guaranteed way to predict whether an invention is novel, nonobvious, patentable, or likely to infringe another's patent. There are two primary limitations on patent searches:

- ***Innumerable Resources.*** Remember that after March 16, 2013, an invention is not novel if it has been patented, described in a patented publication, or in public use, on sale or otherwise available to the public *anywhere in the world* before the filing date of the application (with the grace period previously described). Thus, there are literally millions of sources that might bear upon novelty, including more than eight million issued patents, magazine articles, books, academic papers, websites, and the like. It is not possible for any search to cover all of these resources.

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- ***Pending Patent Applications.*** Patent applications are maintained in secrecy until 18 months after the date the application is filed with the USPTO (and some are kept secret until a patent issues). During this period, it is not possible to obtain any information about the pending application. Thus, it is possible that a review of USPTO records might lead a patent attorney to believe the client's invention is new, nonobvious, and noninfringing, and the attorney would then recommend that the client file a patent application. The USPTO might thereafter publish an existing patent application that would bar the client's application, and there would have been no absolute way to anticipate or predict the existence of such an application.

Patent Opinions

Once the appropriate search has been conducted, the results must be analyzed and communicated to the client. Typically, the results of a professional search company are provided to the law firm, which analyzes all materials and provides a formal legal opinion on patentability of the invention or the likelihood that an invention infringes another invention.

Although patent opinion letters will differ because they will address different issues (e.g., some address patentability, while others address freedom to operate or infringement), they usually include several common elements:

- The letter generally begins by reiterating the client's request for the opinion.
- Opinion letters include a description of the client's invention so that the client can correct any errors in the law firm's understanding of the invention.
- The letter will fully describe the search methodologies used by the firm, including a description of which databases were reviewed, the types of prior art that were examined, whether foreign publications were analyzed, and so forth.
- The letter will identify any relevant patents. Each patent will be fully analyzed and a conclusion will be given as to whether and how each patent located bears upon patentability, infringement, and so forth.
- A formal conclusion will be given indicating whether, in the law firm's professional opinion, the invention is patentable, whether the client has the freedom to operate and bring the invention to market, or whether the invention infringes any patents. The conclusions provided are not phrased as certainties but rather as probabilities. Thus, it is common to see language such as "It is more likely than not" or "Based upon our review, we believe that" and similar somewhat equivocal language.

ETHICS EDGE

AVOIDING THE UNAUTHORIZED PRACTICE OF LAW

Although paralegals are intimately involved in numerous patent-related tasks, they cannot provide legal advice. Thus, avoid giving any advice or opinion to a client regarding patentability, infringement, and so forth. While paralegals play a significant role in conducting searches and may assist in drafting opinion letters, only a licensed attorney can provide legal advice. Thus, an attorney must always sign any opinion letter.

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- The letter will end with disclaimers reminding the client that apparently favorable results from a search do not necessarily guarantee that a patent can be obtained, that due to human error and mistakes in filing it is possible that not all relevant patents were disclosed, that the opinion is not a guarantee, and that the opinion is solely for the client's use and not for the use of others (such as bankers or investors).

THE PATENT APPLICATION PROCESS

Overview of the Application Process

The process of preparing, filing, and shepherding a patent application through the USPTO toward issuance is called "prosecution" (just as is the process of obtaining a trademark registration). An application may be filed by the inventor himself or herself or, as is more usual, by a patent attorney. Only 20 percent of all applications are filed by inventors without the assistance of attorneys. After the application is filed with the USPTO, it will be assigned to one of more than 6,700 patent examiners having experience in the area of technology related to the invention who will review the application and conduct a search of patent records to ensure the application complies with the statutory requirements for patents (including novelty, usefulness, and nonobviousness). Generally, there will be some objection made by the examiner, which will be set forth in a document called an office action. The applicant or attorney typically responds to the office action either by telephone or in writing. If the rejections cannot be overcome, the application may be abandoned or the examiner's refusal may be appealed to the Patent Trial and Appeal Board (until September 16, 2012, called the Board of Patent Appeals and Interferences). Alternatively, the examiner will accept the response(s) to the office action(s) and allow the application. A Notice of Allowance will be sent to the applicant, which specifies an issue fee that must be paid to the USPTO in order for the patent to be granted. Until 2000, all patent applications were maintained in confidence. Since November 2000, however, most patent applications are published 18 months after their filing date (although the applicant can avoid publication by certifying that the invention disclosed in the application has not and will not be the subject of an application filed in another country). Once a patent issues, however, the entire application file (the "file wrapper") becomes a matter of public record. In 2011, patents were granted for about 46 percent of all applications filed.

It generally takes about three years to prosecute a patent (in fiscal year 2011, the average time was 34 months), and costs and fees can range from $5,000 to more than $30,000, with fees generally ranging from $10,000 to $15,000. Because of the time and the high costs involved in obtaining a patent, inventors should conduct a cost-benefit analysis to determine whether the invention merits the effort and expense of obtaining a patent. If the invention will have limited application, it may not be worth securing a patent for it. On the other hand, if the invention is likely to be a commercial success, the time and expense involved in obtaining patent protection will be a worthwhile investment.

A flowchart showing the patent prosecution process is available on the USPTO's website (with links to helpful information) at http://www.uspto.gov/patents/process/index.jsp.

Patent Practice

While preparing trademark and copyright applications is relatively straightforward, preparing a patent application requires skillful drafting as well as knowledge in the relevant field, whether that is biotechnology, chemistry, mechanical engineering, physics, computers, pharmacology, electrical

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engineering, and so forth. Because patent practice is highly technical, law firms that provide patent services to clients generally have a number of attorneys with different skill sets, and the patent department itself may be divided into different groups, such as a mechanical group, a biotech group, and an electrical group. Most **patent attorneys** possess both a law degree and an advanced degree in engineering, physics, chemistry, or the like. Due to the highly skilled nature of patent work, patent attorneys are highly marketable and often command salaries significantly higher than other attorneys. Similarly, other experienced IP professionals are also in high demand.

To represent patent applicants before the USPTO, an attorney must be licensed to practice law in one state and must also be registered to practice with the USPTO. An attorney must pass a registration examination (often called the "patent bar exam"), which requires the attorney to demonstrate he or she possesses the legal, scientific, and technical qualifications to represent patent applications. The examination is a six-hour 100-question multiple-choice test. The examination is computer-based and all questions are drawn from the *Manual of Patent Examining Procedure* and other reference materials. The examination is very difficult, and the pass rate tends to hover around 50 percent.

In addition to the inventor himself or herself and registered patent attorneys, individuals called registered **patent agents** can prosecute patent applications. Patent agents are not attorneys but rather skilled engineers and scientists who take and pass the patent registration examination. All are college graduates or must have the equivalent of such a degree. Although patent agents can engage in patent prosecution, they may not engage in activities that constitute the practice of law, such as representing parties in patent infringement actions or providing legal advice. Many paralegals and law clerks are registered patent agents.

The USPTO website (https://oedci.uspto.gov/OEDCI/query.jsp) provides an index to the thousands of attorney and patent agents who are registered to practice before the USPTO. Many cities also have attorney referral services that can recommend patent attorneys or agents. Finally, a local or state bar association may have associations of patent attorneys and agents.

Confidentiality of Application Process and Publication of Patent Applications

For more than 200 years, all patent applications filed with the USPTO were maintained in strict confidence throughout the entire application process. Only when the patent was issued was the file wrapper open to public inspection. Under the American Inventors Protection Act (AIPA) of 1999, however, which took effect in November 2000, the USPTO now publishes utility and plant applications 18 months after their filing unless the applicant requests otherwise upon filing and certifies that the invention has not and will not be the subject of an application filed in a foreign country that requires 18-month publication. 35 U.S.C § 122. Applicants may request earlier publication. A fee of $300 is charged by the USPTO for publication. If the applicant later decides to apply for a patent in a foreign country that requires 18-month publication, the applicant must provide notice of this foreign filing to the USPTO within 45 days or the application will be regarded as abandoned.

Basic information about the patent is published and the information is accessible through the USPTO's website.

The intent of the 2000 law was to harmonize U.S. patent procedures with those of other countries, almost all of which publish patent applications after an initial period of confidentiality. Most commercial inventors supported the idea of preissuance

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publication because it indicates trends in patents and allows inventors to design their new inventions in ways to avoid infringement. The new act protects inventors from having their published inventions infringed by providing that patentees can obtain reasonable royalties if others make, use, or sell the invention during the period between publication and actual grant of the patent. In any event, once a patent application matures into an issued patent, the entire file wrapper is available for review by the public.

Types of Applications

Although the prosecution of patents is essentially the same for any type of patent application, there are different types of applications. They are as follows:

- ***Provisional application.*** Effective in 1995, and as a result of the adherence of the United States to the **General Agreement on Tariffs and Trade (GATT)**, it is possible to file a **provisional patent application** with the USPTO. A provisional application is less formal than a full utility patent application. It is intended as a relatively inexpensive and expeditious way of embarking on patent protection. A provisional application need not include any claims, and the filing fee is inexpensive ($250 for large entities and $125 for small entities). The applicant must, however, file for a standard utility patent within 12 months of filing the provisional application or the provisional application will be deemed abandoned. A provisional application may be most useful when an inventor is in a race with a competitor and wishes to be the first to file an application. Filing a provisional application allows the inventor to mark the invention with the notice "patent pending." The 20-year term for a utility patent begins with the filing of the actual utility patent. Thus, filing a provisional application and then filing a corresponding nonprovisional application allows an inventor to delay the start of the 20-year period of patent protection for as much as 12 months. Moreover, it provides the inventor a simplified filing with a lower initial investment with one full year to assess the invention's commercial potential before committing to the higher costs of filing and prosecuting a standard utility patent. Although there are obvious advantages to filing a provisional patent application, there are disadvantages as well, primarily that the application will not be reviewed by an examiner during its 12-month pendency. Thus, the inventor is left without any indication regarding the ultimate protection for his or her invention.
- ***Utility application.*** An inventor may file a utility application for a new, useful, and nonobvious process, machine, article of manufacture, or composition of matter or some improvement thereof. If a provisional application has previously been filed by an inventor, a **utility application** must be filed within 12 months of the filing date of the provisional application.
- ***Design application.*** A **design application** seeks protection for new, original, and ornamental designs for articles of manufacture.
- ***Plant application.*** A **plant application** seeks protection for new and distinctive asexually reproduced plants.
- ***Continuing application.*** A **continuing application** claims priority from a previously filed application. There are two types of continuing applications: continuation applications and continuation-in-part applications. The **continuation application** is a continuation patent filed when an examiner issues a final office action rejecting some claims in an application, and the inventor wishes to proceed with the allowed claims and then continue to pursue the rejected claims in a separate application. The continuation application must be filed before

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the original application is either abandoned or granted. The applicant and the patent examiner continue to discuss the rejected claims while the approved claims go forward toward issuance of a patent. The filing date for the continuation application is the same as the filing date of the original earlier-filed **parent application**. Thus, its effect is that it may shorten the term of existence of the later-issued patent (because the term of a patent is 20 years from the filing date of the original or parent application). On the other hand, by capturing the date of its parent, it may circumvent recent prior art. The other type of continuing application is the **continuation-in-part (CIP)** application, which contains significant matter in common with the original or "parent" application and also adds new matter not disclosed in the earlier parent, usually because an improvement to the invention was developed subsequent to the filing of the parent application. Claims that relate to the later-filed CIP (e.g., the newly added matter) are entitled to the filing date of the CIP rather than the filing date of the original application. Claims that relate to the original application retain the original filing date.

- ***Divisional application.*** After an application is filed, the examiner may determine that the application covers more than one invention. Because each patent application can cover only one invention, a **divisional application** will be created to carve out a new application for the additional invention. The new divisional application typically retains the filing date of the original or parent application. The USPTO refers to divisional applications as a type of continuing application.
- ***PCT application.*** A **Patent Cooperation Treaty (PCT)** application allows an applicant to file one application that may be relied upon for later filing in countries that are members of the Patent Cooperation Treaty, to which more than 140 nations belong, including the United States. **PCT applications** are administered by the World Intellectual Property Organization (WIPO). Individual filings in each foreign country in which patent protection is desired would be an extremely expensive process inasmuch as the inventor might discover there is no market for the invention in some countries and might wish to abandon the process. The PCT application process allows an inventor to file one standardized application and then postpone prosecuting the applications in other member nations for 30 months. The filing date for each foreign country is deemed to be the date identified in the original application. Although often called an "international application," the application does not automatically result in registration in member nations of the PCT. The application must still be prosecuted separately in each individual nation in which the inventor desires protection. The primary purpose of a PCT application is to allow an inventor to delay prosecution in foreign countries until a determination is made about whether protection is desirable in those countries. PCT applications are discussed in Chapter 21.

Preparing the Application

Except where noted, the following discussion relates to regular utility patent applications rather than to provisional applications or to design, plant, or other patent applications. An application for patent must be in English (or accompanied by an English translation), and it includes the following elements (which are shown later in Exhibit 18–6, an issued patent):

- A **specification** (the part of the application that describes the invention and the manner and process of making and using it) and the **claims** (separate paragraphs that distinctly claim the

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subject matter that the applicant regards as the invention)
- A drawing (when necessary for understanding of the invention)
- An oath by the applicant stating the applicant's belief that he or she is the original and first inventor of the invention

A filing fee must also accompany the application. Although these elements are few, their importance cannot be underestimated. Drafting the specification (describing the invention) is difficult and painstaking work. For a complex invention, it may take 40 hours or more. The level of detail and specificity imposed on applicants is a tradeoff: In return for obtaining a 20-year monopoly during which they may exclude others from selling, making, or using their inventions, inventors must fully disclose to the public what the invention is and exactly how it works.

Specification. The **specification** is the part of the application that describes the invention and the manner and process of making it and using it. It must be made in "such full, clear, concise, and exact terms as to enable any person skilled in the art . . . to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention." 35 U.S.C. § 112.

Section 112 requires that the specification must be so complete as to enable one skilled in the art to make and use the invention. Thus, a general description such as "the invention is a high-speed drill" will be insufficient. In many instances, an application is rejected because of a **nonenabling specification**, meaning the specification is not sufficient to teach or enable another to make or use the invention. Simply put, stating what the invention is is insufficient; the application must describe how the invention works. A specification may incorporate by reference other materials such as pending or issued patents, publications, and other documents as part of the enabling requirement.

In addition to the enabling requirement, the application must set forth the "best mode" or preferred method contemplated by the inventor for making the invention. When the patent ultimately expires, competitors will then be able to compete with the former patentee on an equal footing because they will know how to make and use the invention.

The specification itself is composed of several distinct elements:

- ***Title.*** The invention must be given a short and specific title that appears as a heading on the first page of the application. Examples of titles are "Bat for Baseball," "Nail Gun," and "Combined Refrigerator and Microwave Oven with Timed Overload Protection."
- ***Cross-references to related applications.*** If the application seeks the benefit of the filing date of a prior invention applied for by the same inventor or claims an invention disclosed in an earlier application by the same inventor, the second application must provide a cross-reference to the earlier application and identify it by serial number and filing date. For example, a continuing application must identify its parent application.
- ***Background.*** The **background** section of the specification should identify the field of the invention and discuss how the present invention differs from the known art. Generally, this section critiques other inventions and demonstrates the need and worth of the invention being applied for.
- ***Brief summary of invention.*** The summary section provides a short and general statement of the nature, operation, and substance of the invention. It may point out advantages of the invention and how it solves existing problems.
- ***Brief description of drawings.*** If drawings are included, they should be briefly described. The applicant must include a listing of all figures by

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number (e.g., Figure 1-A) with corresponding statements explaining what each figure depicts.

- ***Detailed description of the invention.*** In this section, the invention must be explained along with the process of making and using the invention in full, clear, concise, and exact terms. This description must be sufficient so that any person of ordinary skill in the pertinent art or science could make and use the invention without extensive experimentation. The best mode contemplated by the inventor of carrying out the invention must also be set forth. Each element of the drawing should be mentioned. This section was previously called "Description of the Preferred Embodiments."
- ***Claims.*** A specification must include a least one **claim**. The claims define the scope of the invention. Although the language is precise and each claim is limited to one sentence, the inventor's goal is to draft the claims in such a way as to achieve the broadest possible scope of protection for the invention and yet comply with the statutory requirement of specificity. The claims are the most significant part of the application. Whether a patent is granted is determined in large measure by the choice of words of the claims. The claims will be compared against the prior art to determine whether the invention is entitled to be patented. If the inventor later alleges another has infringed his or her invention, a court will compare the claims set forth in the patent with the alleged infringer's invention in determining whether infringement has occurred. Moreover, the filing fee required is determined in part by the number of claims. Claims are often compared to the descriptions of real estate found in deeds that describe the "metes and bounds" of a parcel of property. A patent claim similarly describes the boundaries of the claimed invention. Although there is no statutory form for the claims, the USPTO practice is to prefer that each claim be the object of a sentence starting with "I (or "we") claim . . ." or "What is claimed is . . . ." Each claim begins with a capital letter and ends with a period, and they should be arranged in order of scope so that the first claim is the least restrictive. Moreover, claims have a set structure of three basic elements. They begin with an introductory phrase, called a *preamble*, that provides a general description of the invention. The preamble is followed by a *transitional phrase* or statement, such as "comprising" or "consisting of," and the transitional phrase is then followed by the *body* of the claim, which identifies the elements or steps of the invention. Thus, for example, a claim might read as follows: "The invention claimed is a seat belt system comprising a seat belt, a retractor capable of retracting the seat belt, and a controller . . . ." There are different types or formats of claims:
    - ***Independent claims.*** *Independent claims* describe the invention in a general and broad manner. They stand by themselves and do not refer to any other claims.
    - ***Dependent claims.*** *Dependent claims* are more narrowly stated and refer back to or incorporate all elements of the more broadly stated independent claims. Dependent claims cannot stand on their own; they must be read with one or more of the previously stated claims. For example, a dependent claim might read as follows: "The exercise device of Claim 1, wherein the apparatus is a circular trampoline." Thus, each dependent claim is narrower and more specific than any preceding claim. Applicants may also use multiple dependent claims, which are claims that refer back in the alternative to more than one preceding independent or dependent claim (for example, "a machine according to Claims 3 or 4,

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further comprising . . . ."). Examples of independent and dependent claims can be seen in Exhibit 18–1.

- ***Functional claims.*** *Functional claims* define the invention by what it does rather than in terms of its structure. These types of claims are also referred to as "means-plus-function" claims or "means claims" and might read as follows: "A means for attaching the gadget described in Claim 1 to the panel described in Claim 2."
- ***Product-by-process claims.*** *Product-by-process claims* define a product by its process of preparation or manufacture and are often seen in chemical or pharmaceutical inventions. A product-by-process claim might read as follows: "A synthetic steel material prepared by a process comprising the steps of . . . ."
- ***Jepson claims. Jepson claims*** (named for the inventor who used this format) are used for improvements to existing inventions and identify what is new to the invention. They are useful for pointing out to the patent examiner's attention the specific novelty of an invention and might read as follows: "A locking fuel pump dispenser nozzle having a nozzle connected to a fuel pump, the improvement comprising a locking mechanism . . . ."
- ***Markush claims.*** *Markush claims* (also named after an inventor and sometimes called Markush Groups) are found in inventions relating to chemicals. A Markush claim recites alternatives in a format such as "An acid inhibitor selected from the group consisting of A, B, and C."

Design patent applications have only one claim—for example, "the ornamental design of a child's chair, as shown and described." Plant patents also have only one claim inasmuch as the entire plant is claimed as the inventive material. (See Exhibit 18–1 for a sample of claims.)

- ***Abstract of the disclosure.*** Strictly speaking, the abstract is not part of the specification but is rather a concise statement of the invention. The purpose of the abstract is to enable the USPTO and the public to determine quickly the nature of the technical disclosures of the invention. It points out what is new in the art to which the invention pertains. The abstract begins on a separate page and should not be longer than 150 words. The **abstract** should be a single short paragraph. The abstract for U.S. Patent No. 6,929,573 entitled "Bat for Baseball" is as follows:

  > An improved baseball bat comprises a tubular core of rigid materials having a first section to support a handle and a second section to support a striking portion. The second section of the core has a diameter relatively larger than that of the first section of the core. A covering member of semirigid materials has a handle portion embracing the first section of the core and a striking portion embracing the second section of the core. Whereby, the baseball bat has a light weight, good equilibrium, and high structural strength for a good performance.

  An abstract for a plant patent might read, "A hybrid tea rose having two-toned blossoms of pink with a white reverse."

Drawings, Models, and Specimens. If drawings are needed to understand the invention, they must be included. Nearly all patent applications are accompanied by drawings. The drawings must also be described so that the viewer knows whether the drawing is a cross-section, a side view, and so

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WHAT IS CLAIMED IS:

1. An improved baseball bat comprising: a tubular core of rigid materials having a first section to support a handle and a second section to support a striking portion, said second section having a diameter relatively larger than that of said first section; a covering member of semirigid materials having a handle portion embracing said first section of said core and a striking portion embracing said second section of said core; wherein said first section and second section of said core are respectively made and connected by a connecting means; and wherein said connecting means is a ring-like device inserted tightly into one end of said second section of said core.
2. The baseball bat as claimed in claim 1, wherein said core is made of a material selected from a group consisting of composite materials, metals, and plastics.
3. The baseball bat as claimed in claim 2, wherein said core is made of fiber-reinforced plastic materials.
4. The baseball bat as claimed in claim 1, wherein said handle portion of said covering member is made of foam plastic materials and said striking portion of said cover is made of wood materials.
5. The baseball bat as claimed in claim 1, wherein the end of said second section of said core has an inner shoulder to complementedly connect with an outer shoulder formed on said ring-like device.
6. The baseball bat as claimed in claim 1, wherein said second section of said core is taperedly formed.
7. The baseball bat as claimed in claim 1, further comprising a protecting layer made of fiber-reinforced material and wrapping around the surface of said striking portion of said covering member.
8. The baseball bat as claimed in claim 1, further comprising at least one shock-absorbing device respectively and tightly inserted inside said core.
9. The baseball bat as claimed in claim 1, further comprising at least one weight device respectively and tightly inserted inside said core.
10. The baseball bat as claimed in claim 2, wherein the end of said second section of said core has an inner shoulder to complementedly connect with an outer shoulder formed on said ring-like device.
11. The baseball bat as claimed in claim 3, wherein the end of said second section of said core has an inner shoulder to complementedly connect with an outer shoulder formed on said ring-like device.
12. The baseball bat as claimed in claim 4, wherein the end of said second section of said core has an inner shoulder to complementedly connect with an outer shoulder formed on said ring-like device.

EXHIBIT 18–1 Claims for Utility Patent (U.S. Patent No. 6,929,573)

forth. For example, the drawings might be described as follows: "Figure 3 is a right-hand perspective view of the child's chair. Figure 2 is a front plan view thereof." The USPTO has stringent requirements relating to the size, symbols, and format of drawings, and usually a graphic artist or patent draftsperson is retained to prepare the drawings. Many law firms that practice patent law employ draftspersons on a full-time basis. Using new computer-assisted design and drawing software, many patent practitioners can prepare their own drawings. Informal drawings (photocopies) may be submitted if they are legible, but the examiner will usually impose a requirement that formal pen-and-ink drawings be submitted. The drawings must show every feature of the invention as specified in the claims.

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Applicants were formerly required to submit a working model of the invention, but this requirement was eliminated in the late 1800s (although an examiner does have the authority to require that a working model be submitted). If the invention relates to a composition of matter, the USPTO may require the applicant to furnish specimens or ingredients for the purpose of inspection or experiment.

Oath of Inventor. The applicant must sign an oath or declaration that he or she believes himself or herself to be the original and first inventor of the process, machine, manufacture, or composition of matter, or improvement thereof. The applicant must also identify his or her country of citizenship and provide an address. The applicant must acknowledge that he or she has reviewed and understands the contents of the application and understands the duty to disclose all information known to be relevant to patentability. An oath is made before a notary public, while a declaration is a statement by the applicant acknowledging that willful false statements are punishable by law. Either an oath or a declaration is acceptable.

Effective September 16, 2012, under the AIA, an applicant may submit a "substitute statement" in lieu of an oath or declaration if the inventor is deceased, unable to do so, or unwilling to do so and is under an obligation to assign the invention. Thus, an employer could file the patent application without an oath from the individual inventor if the inventor is under an obligation to assign the invention to the employer and is unwilling to make the oath. An individual who is under an obligation to assign the invention may include the substitute statement in the assignment document and this will be sufficient. These new provisions simplify the application process for employers who own the rights to their employees' inventions and avoid the problems associated with uncooperative or unavailable inventors.

For more than 200 years, the Patent Act required that the person claiming the patent must be the actual inventor. 35 U.S.C. § 111. The applicant could not be an assignee of the inventor. Thus, a review of the more than eight million applications on file with the USPTO through September 16, 2012, will disclose that in nearly every case, the application has been made by an individual. Applications were not made by General Electric, Ford Motor Company, Microsoft Corporation, or any other corporation or business entity. Although these companies may employ the inventors and may, in fact, own the invention (due to agreement between the parties that the employer will own any invention made by the employee), the application itself must have been signed by the individual inventor. The rationale for this law was that it would help ensure that individual inventors would not have their inventions stolen.

Effective September 16, 2012, under the AIA, a person to whom the inventor has assigned the invention or has an obligation to assign the invention may apply for the patent. This new provision will thus allow employers such as Microsoft to file patent applications when their employees have assigned or are under an obligation to assign their inventions to them, streamlining the application process for large employers who may have lost track of their employees and cannot locate them in order to have them file patent applications. These new provisions of the AIA make it easier for the actual owners of inventions to file applications.

Subject to the above new provisions of the AIA, if an invention has been made by two or more persons jointly, they must file the application jointly. A **joint application** may be made even if the inventors did not physically work together or at the same time, did not each make the same type or amount of contribution, or did not each make a contribution to the subject matter of every claim in the patent. Accurate records should be kept regarding each inventor's contributions so that if one inventor's contributions

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are refused or dropped from the claims he or she can be removed from the application.

An application can be assigned to another person or company concurrently with the execution of a patent application or at any time thereafter. Concurrent assignments have taken place when companies own the inventions created by their employees. The new provisions of the AIA allowing patent owners such as employers to file patent applications will likely eliminate the practice of concurrent assignments. Nevertheless, all assignments should be recorded with the USPTO to provide public notice of the owner of the patent rights even though recordation is not required to make an assignment valid. Issues relating to joint inventors, patent ownership, and transfer are further discussed in Chapter 19.

If anyone other than the actual inventor or patent owner is filing the application, a power of attorney will be needed to authorize the patent attorney or patent agent to act on the inventor's behalf. The power of attorney may be a separate form or may be included as part of the oath or declaration. The USPTO supplies a form for powers of attorney.

The requirements for preparing and filing design and plant patents are nearly identical to those discussed earlier for utility patents. Plant patent applications, however, require the oath or declaration to confirm that the plant was reproduced asexually. Design and plant patent applications include only one claim.

Filing the Application. After the application has been thoroughly reviewed to ensure it complies with USPTO regulations (such as requirements relating to the size of the paper used and that all pages be numbered and one and one-half or double-spaced, if one files a paper application rather than filing electronically), the application package should be assembled for filing with the USPTO.

Although patent applications may be submitted by mail to the USPTO, the vast majority of applications are filed electronically, using the USPTO's **electronic filing system** called "EFS-Web," which was implemented in 2000. As of 2011, approximately 93 percent of all patent applications were filed using EFS-Web. The system uses standard Web-based screens and prompts to enable filers to submit PDF documents directly to the USPTO. A key component of EFS-Web is the use of PDF fillable-form documents, which are interactive forms with various field types and formatting options that auto-load information directly into the USPTO systems. Filers may submit documents as either nonregistered or registered users. While anyone may file a patent application electronically with the USPTO, most practitioners and frequent filers become "registered users" by applying for a customer number and digital certificate (which uniquely identifies the user and allows secure access to the user's patent data). When the application (and fee) are submitted electronically, the USPTO issues an electronic receipt and confirmation.

A transmittal form should be prepared (the USPTO provides a fillable form—see Exhibit 18–2), and it informs the USPTO as to what is being filed, identifies the applicant, and indicates if any other documents accompany the application. The applicant typically checks boxes indicating that a specification is included, an oath or declaration is included, and so forth. If the application is being simultaneously assigned at the same time it is filed, a recordation form should be included and the appropriate box should be checked on the application transmittal form. (The recordation form used for patent assignments is nearly identical to that used for trademarks shown in Chapter 5 as Exhibit 5–4.)

Applicants may also submit an Application Data Sheet, a fillable form supplied by the USPTO that contains bibliographic data such as information about the applicant, correspondence information, and information as to whether the applicant is represented by a patent attorney or agent. Use of the

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UTILITY PATENT APPLICATION TRANSMITTAL (Only for new nonprovisional applications under 37 CFR 1.53(b))		
	Attorney Docket No.	
	First Inventor	
	Title	
	Express Mail Label No.	

APPLICATION ELEMENTS
See MPEP chapter 600 concerning utility patent application contents.

ADDRESS TO: **Commissioner for Patents, P.O. Box 1450, Alexandria VA 22313-1450**

1. ☐ **Fee Transmittal Form** (e.g., PTO/SB/17)
2. ☐ **Applicant claims small entity status.** See 37 CFR 1.27.
3. ☐ **Specification** [*Total Pages*__________]
   Both the claims and abstract must start on a new page
   (*For information on the preferred arrangement, see MPEP 608.01(a)*)
4. ☐ **Drawing(s)** (*35 U.S.C. 113*) [*Total Sheets* _________]
5. **Oath or Declaration** [*Total Sheets* _________]
   a. ☐ Newly executed (original or copy)
   b. ☐ A copy from a prior application (37 CFR 1.63(d))
   (*for continuation/divisional with Box 18 completed*)
   i. ☐ DELETION OF INVENTOR(S)
   Signed statement attached deleting inventor(s) name in the prior application, see 37 CFR 1.63(d)(2) and 1.33(b).
6. ☐ **Application Data Sheet.** See 37 CFR 1.76
7. ☐ **CD-ROM or CD-R** in duplicate, large table or Computer Program (*Appendix*)
   ☐ Landscape Table on CD
8. **Nucleotide and/or Amino Acid Sequence Submission**
   (*if applicable, items a. – c. are required*)
   a. ☐ Computer Readable Form (CRF)
   b. Specification Sequence Listing on:
   i. ☐ CD-ROM or CD-R (2 copies); or
   ii. ☐ Paper
   c. ☐ Statements verifying identity of above copies

ACCOMPANYING APPLICATION PARTS

9. ☐ **Assignment Papers** (cover sheet & document(s))
   Name of Assignee________________________
10. ☐ **37 CFR 3.73(b) Statement** (*when there is an assignee*) ☐ **Power of Attorney**
11. ☐ **English Translation Document** (*if applicable*)
12. ☐ **Information Disclosure Statement** (PTO/SB/08 or PTO-1449)
    ☐ Copies of citations attached
13. ☐ **Preliminary Amendment**
14. ☐ **Return Receipt Postcard** (MPEP 503)
    (*Should be specifically itemized*)
15. ☐ **Certified Copy of Priority Document(s)**
    (*if foreign priority is claimed*)
16. ☐ **Nonpublication Request** under 35 U.S.C. 122(b)(2)(B)(i). Applicant must attach form PTO/SB/35 or equivalent.
17. ☐ Other:________________________

18. If a CONTINUING APPLICATION, *check appropriate box, and supply the requisite information below and in the first sentence of the specification following the title, or in an Application Data Sheet under 37 CFR 1.76:*

☐ Continuation ☐ Divisional ☐ Continuation-in-part (CIP) of prior application No.: ……………………………

Prior application information: *Examiner* ________________ *Art Unit:* ________________

19. CORRESPONDENCE ADDRESS

☐ The address associated with Customer Number: [ ] ***OR*** ☐ Correspondence address below

Name					
Address					
City		State		Zip Code	
Country		Telephone		Email	

Signature		Date	
Name (Print/Type)		Registration No. (Attorney/Agent)	

This collection of information is required by 37 CFR 1.53(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**
If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

EXHIBIT 18–2 Utility Patent Application Transmittal
Source: http://www.uspto.gov/forms/sb0005_fill.pdf

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Although the AIA made significant changes to patent law, it did not include some changes that many had hoped for. For example, it did not prevent "forum shopping" by plaintiffs in patent infringement cases. At present, the Eastern District of Texas is perceived to be plaintiff-friendly, and more defendants are sued there for patent infringement than in Delaware, California, New Jersey, and Illinois combined. Similarly, the AIA did not address proposals to rein in patent infringement damage awards. Finally, it did not halt the practice of diverting USPTO fees to other government agencies. Most experts believe that these proposals, which had been hotly contested for years, were dropped from the final text of the AIA to ensure that some meaningful patent reform could go forward.

INTRODUCTION TO INTERNATIONAL PATENT PROTECTION

The rights granted by a U.S. patent extend only throughout the United States and have no effect in a foreign country. Therefore, generally, an inventor who desires patent protection in other countries must apply for a patent in each of the other countries or in regional patent offices. Nearly every country has its own patent law, and a person who wishes to obtain a patent in a particular country must make an application for patent in that country, in accordance with its requirements. A directory of and links to worldwide patent offices can be found at http://www.wipo.int/directory/en/urls.jsp.

The laws of many other countries differ in various respects from the patent law of the United States. In many foreign countries, publication of the invention before the date of the application will be an absolute bar to the right to a patent. In the United States, however, effective under the AIA on March 16, 2013, the one-year grace period applies as to the inventor's own publication or disclosure, so that if the invention was described by the inventor (or one who obtained the invention from the inventor) in a printed publication, the inventor has one year thereafter to file the patent application for the invention. Most foreign countries require that the invention be manufactured in that country within a certain period of time, usually three years, after grant of the patent or the patent will be void, while in the United States there is no requirement that the invention ever be manufactured, used, or sold. Additionally, nearly all foreign countries grant patents to the "first to file" the application. The United States has followed a "first to invent" system for 200 years, although under the AIA, effective March 16, 2013, the United States will adopt a "first to file" system.

There are several **international patent** treaties to which the United States adheres, primarily the Paris Convention, the PCT, and the Agreement on TRIPS.

THE PARIS CONVENTION

The Paris Convention for the Protection of Intellectual Property of 1883 is a treaty adhered to by more than 170 nations and is administered by WIPO, discussed in Chapter 8.

The Paris Convention requires that each member country guarantee to the citizens of the other member adherents the same rights in patent and trademark matters that it provides to its own citizens (the principle of "national treatment"). The treaty also provides for the right of priority in the case of patents, trademarks, and industrial designs (design patents). The right of priority means that, on the basis of a patent application filed in one of the member countries, the applicant may, within one year, apply for patent protection in any of the other member countries. These later applications will then

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be regarded as if they had been filed on the same day as the first application in the first country, assuming the first application adequately disclosed the invention. (See Appendix A for a table of the countries adhering to the Paris Convention.)

Recall from Chapter 8 that after a trademark application is filed in the United States (or any Paris Convention member country), the applicant has six months to file an application for the same mark in any Paris Convention member country. The later-filed application captures the filing date, called the "priority date," of the earlier-filed application. The same principle is true for patents, although the time period for filing a patent application is one year (six months for a design patent). Thus, a later application will have priority over an application for the same invention that may have been filed during the 12-month period of time. For example, if an inventor files a patent application in the United States on January 1, 2012, he or she will have until January 1, 2013, to file an application for the same invention in any Paris Convention member country, which application will then be treated as if it were filed in that country on January 1, 2012. Filing either a provisional application or a standard utility patent begins the Paris Convention priority year for U.S. applicants.

Additionally, the later-filed application, because it is based on the date of the first application, will not be invalidated by some act accomplished in the interval, such as sale or use of the invention. Similarly, the earlier priority date is the date of invention for determining whether prior art precludes granting of a patent for the invention.

There are, however, a few conditions that must be satisfied if an applicant is to be allowed to claim the date he or she first filed an application in a foreign country as the priority date in a later-filed U.S. application:

- The foreign country in which the application was first filed must afford similar privileges to citizens of the United States or to applications first filed in the United States.
- The applicant must submit a certified copy of the original foreign application, specification, and drawings to the USPTO.
- The priority application must be for an invention by the same inventor(s) and for the same invention as identified in the later U.S. application.

The priority right is based on the filing of the application in the foreign country and timely filing in the United States. The prosecution status or history of the application in the foreign country is irrelevant. If the foreign country refuses to grant a patent, or the applicant abandons the application, such is irrelevant to the later U.S. application. Similarly, the granting of a patent by one country does not oblige another country to do so.

The Paris Convention affords patent applicants the opportunity to file a patent application in a member country and then take 12 months to determine whether foreign protection will be sought. An applicant may determine that the invention is not marketable in certain countries and thus decide not to apply for patents in those countries. In this way, the Paris Convention saves an applicant the time and expense of having to file simultaneous patent applications in several nations before it has had an opportunity to evaluate the likelihood of obtaining patent protection and exploiting the invention commercially.

THE PATENT COOPERATION TREATY

Introduction to the Patent Cooperation Treaty

While the Paris Convention allows applicants to defer decisions about filing in member countries for 12 months, it still requires that applicants file separate applications in each country in which they desire

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protection. For an inventor who wishes to market his or her invention on a global basis, this process is time-consuming and expensive in the extreme. The PCT, which was negotiated in 1970 and came into force in 1978, responds to these concerns by providing a centralized way of filing, searching, and examining patent applications in several countries simultaneously. Moreover, a standardized application format is used, saving applicants substantial time and money that is ordinarily incurred in ensuring that a patent application complies with the procedural and formatting requirements imposed by each country. The PCT is adhered to by more than 140 countries (called "contracting states"), including the United States, and is administered by WIPO. In sum, the PCT allows an inventor to file one "international" application and seek protection for the invention simultaneously in several countries. (See Appendix A for a table of the countries adhering to the PCT.)

The one application filed with the PCT does not automatically mature into a patent that affords patent rights in several countries. The applicant must eventually prosecute the application in the countries elected in the **national phase**. The PCT process, however, affords a significant window of time for the applicant to keep his or her options open while a determination is made as to whether protection should be sought in various foreign countries. The USPTO website (http://www.uspto.gov) provides a great deal of information about the PCT process, including links to a list of PCT member countries, fee schedules and PCT fees and forms, and tutorials relating to the PCT application process.

Phases in the PCT Application Process

There are two main "phases" for PCT applications: the "international phase," which begins with filing the application and includes an international search report and written opinion and that may consist of two parts or "chapters," mandatory Chapter I and an optional Chapter II (collectively referred to as the "international phase"); and the "national phase," which involves prosecution of the application in each country in which the applicant desires patent protection.

The PCT Application Process

Filing the Application.

The "international" application may be filed with the patent office of the member country of which the applicant is a national or resident or, if the applicant desires, with the International Bureau of WIPO in Geneva, Switzerland. When filed with a national patent office, such as the USPTO, the office is said to act as a PCT **receiving office**. Typically, applicants file their PCT applications with their own national patent offices. Thus, the USPTO acts as a receiving office for most international applications filed by nationals or residents of the United States, and the application may be filed electronically using EFS-Web (which provides an instant serial number and increases the visibility of the application on Private PAIR). Effective January 1, 2004, the filing of an international application automatically constitutes the designation of *all* contracting countries to the PCT on that filing date, meaning that the application is viewed as requesting patent protection in the more than 140 PCT countries.

Ultimately, the applicant will designate or elect those countries in which the applicant desires patent protection. For example, a citizen of the United States could file a PCT application with the USPTO and later elect Spain, Brazil, and the United Kingdom as countries in which he or she also desires to pursue patent protection. Only one filing fee is paid for filing the PCT application, which is called an "international patent application." The amount of the filing fee generally depends upon the length of the application. The application is similar in form to utility patent applications in that it contains claims and drawings of the invention.

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A PCT application may claim priority, under the Paris Convention, of an earlier patent application for the invention. Thus, if a patent application was filed in the United States on January 4, 2012, and an international PCT application was filed with the USPTO on June 4, 2012, which later designates or is prosecuted in Spain, Brazil, and the United Kingdom, the effective filing date for the PCT application for all of those countries will be January 4, 2012. If priority is not claimed under the Paris Convention (usually because no prior application has been filed in any foreign country), the effective filing date will be the date the PCT application was actually filed.

Many applicants file a standard patent application with their home patent office and then, near the end of the 12-month period afforded by the Paris Convention, file a PCT application with their home office (now acting as a receiving office).

Chapter I. The filing of the international patent application triggers the first phase of the PCT process, called **Chapter I**. During Chapter I, the international application is subjected to an international search by an "international searching authority," which is one of the experienced patent offices designated by WIPO to conduct searches. The international searching authorities designated by WIPO are the national offices of Australia, Austria, Brazil, Canada, China, Finland, Japan, the Republic of Korea, the Russian Federation, Spain, Sweden, the United States, the Nordic Patent Institute, and the European Patent Office. The applicant typically selects the international searching authority it desires, and it may be different from the office that is serving as the receiving office. The international searching authority will conduct an extensive search of the relevant prior art. The results of the search are set forth in an "international search report" that is provided to the applicant four or five months after the international application is filed (which is also 16 months after the Paris Convention priority date). The search report typically lists and identifies documents and references that may affect patentability. Additionally, the international searching authority will provide a written opinion to the applicant and to WIPO. The opinion is a preliminary (and nonbinding) determination as to whether the invention is patentable. The applicant may amend claims in the international patent application, if necessary to avoid prior art. The applicant is generally given two months to amend his or her claims. If the report discloses prior art that would bar the application, the applicant may decide to abandon the PCT application.

Since January 2009, the applicant may also request a supplementary international search from an international search authority other than one that carried out the initial search. Only 41 supplementary requests were made in 2010.

The PCT application and the international search report are published 18 months after the filing date or priority date. Publication serves to notify the public that an international patent application has been filed for the invention and affords an opportunity for third parties to obtain copies of the application. After April 1, 2006, publication is solely in electronic form (and is available through WIPO's website at http://www.wipo.int). The written opinion is not published and is not publicly available until 30 months from the priority date. The publication date should be docketed so that the inventor can ensure applications are timely filed in countries that are not members of the Paris Convention inasmuch as this publication will foreclose patent applications in those countries that bar applications for inventions that have been published (unless they afford a one-year grace period for the inventor's disclosures, such as the United States does, effective under the AIA). Until publication, the application is maintained in confidence. In a later infringement case, in most PCT countries, a patent owner may recover damages arising from the date of publication (rather than from the date the patent issues).

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The Chapter I phase of the PCT process lasts for 20 months from either the filing date of the international patent application with the receiving office or the claimed priority date if the application claims priority under the Paris Convention. This 20-month period affords the inventor the opportunity to evaluate the marketability of the invention and gather funds in order to enter foreign markets.

Participation in Chapter I of the PCT process does not require participation in Chapter II.

Chapter II. After completion of Chapter I, the applicant may now elect to prosecute the application in individual countries in which patent protection is desired (the "national phase") or may delay prosecution in the national phase until 30 months (in most countries) from the effective filing date of the PCT application. Alternatively, the applicant may take the optional step of entering **Chapter II** and requesting or demanding an international preliminary examination. This demand is generally filed 22 months after the priority date or filing date and identifies or elects the countries in which protection is desired. Filing fees are required. An international preliminary examining authority (the same as the international searching authorities identified earlier) will issue an international preliminary report on patentability (and stating whether the claims satisfy the criteria of novelty, nonobviousness, and industrial application) and communicate it to the various national offices in which the applicant desires patent protection. Although the report is not binding on any specific nation, it is highly authoritative. If the report is favorable, it provides a strong basis on which to continue with the application in various countries. If the report is unfavorable, the applicant may modify his or her claims or decide not to proceed further.

The PCT has experienced a consistent growth rate each year, which growth has generated tremendous increases in the workload for various PCT offices. In many cases, applicants entered Chapter II and demanded an international preliminary examination merely for the purpose of buying time, namely, an additional 10 months while the examination was conducted. As a result, the PCT was modified effective April 1, 2002, to provide that the time limit for entering the national phase (and prosecuting the individual patent applications in the desired countries) will be 30 months from the effective date of filing of the PCT application whether the applicant enters after Chapter I (and forgo an international preliminary examination) or after Chapter II (and demands the international preliminary examination). Thus, applicants who wish to buy time no longer need to file a demand for the international preliminary examination. This new modification has resulted in fewer demands for international preliminary examinations, which in turn reduces the workload for the various patent offices.

Although the United States has adopted the modified rule, not all PCT contracting states have changed their national laws to adopt this new 30-month period.

The National Phase. If, after the duration of Chapter I (and the international preliminary report on patentability, if elected through Chapter II), the applicant decides to go forward with the application in the countries designated in the application, the applicant commences the "national phase" of the PCT application process.

Remember that as of April 1, 2002, an applicant must enter the national phase and begin prosecuting the application in individual countries 30 months from the date of filing of the PCT application or the priority date, whether the applicant enters the national phase after Chapter I (and forgo an international preliminary examination) or after Chapter II (and demands the international preliminary examination). The additional time afforded by the PCT process (30 months from the

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date of filing of the PCT application or its priority date, whether or not an international preliminary report on patentability is requested in Chapter II) is significantly more time (i.e., 18 months more) than the 12 months afforded under the Paris Convention. This additional time is useful for allowing inventors to determine whether the invention is commercially exploitable in various countries and whether protection is needed in certain countries. Note, however, that an applicant may always request early entry into the national phase.

National fees must be paid to each country in which protection is desired, and often translations must be obtained of the PCT application. In many instances, the applicant will decide to forgo protection in some countries and will not pursue the patent application in those countries. Thus, the application will lapse in those countries. Each national office in which the application is pursued will now conduct its own search and examination procedure, although the process is both easier and faster due to the fact that the highly credible international search report and written opinion from Chapter I and possibly an international preliminary report on patentability from Chapter II have already provided interpretations regarding the patentability of the invention. Each of the countries will either grant or reject the application based on their own laws. The term of a patent designating the United States is 20 years from the effective filing date of the PCT application.

See Exhibit 21–1 for a PCT timeline and Exhibit 21–2 for a summary of the PCT application process.

Advantages of the PCT Application Process

The most significant advantage provided by the PCT application process is time. If an application is filed in the United States and the applicant wishes to file an application for the same invention in Japan, under the Paris Convention, the applicant has only 12 months from the U.S. filing date to file in Japan. The filing in Japan will require filing fees, translations,

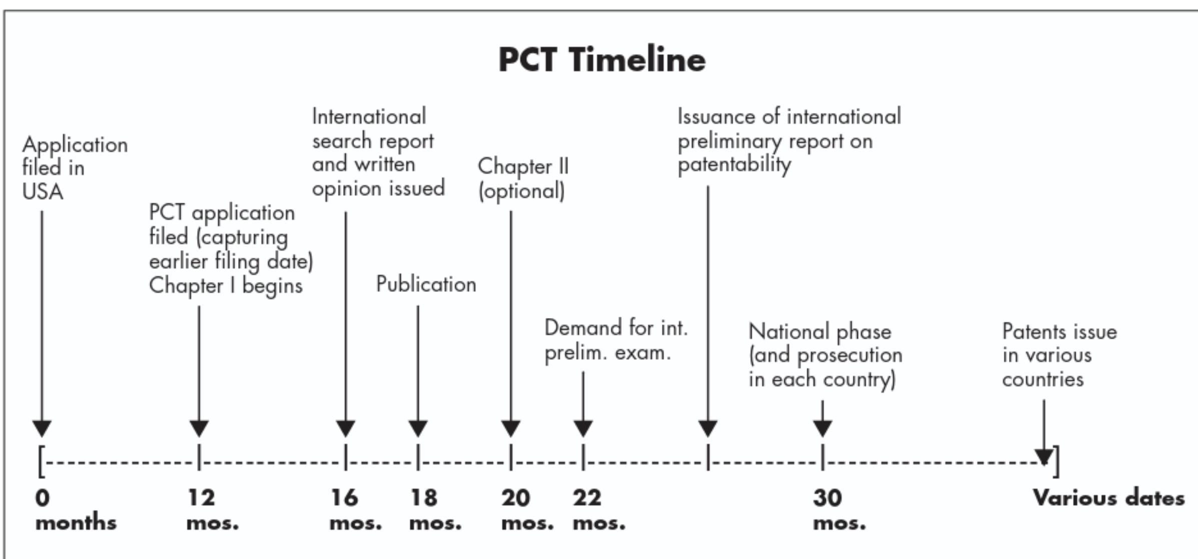

EXHIBIT 21–1 PCT Timeline

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THE PCT PROCESS IN A NUTSHELL

1. **Filing.** An application is filed in a standardized form in a "Receiving Office." Since 2004, the application is deemed to constitute a designation of all PCT contracting states. Filing the application commences mandatory Chapter I of the PCT, which includes an international search and examination. Priority may be claimed under the Paris Convention.
2. **Examination.** The application is examined by an international search authority, which will issue an international search report and a written opinion. The applicant may request a supplementary international search.
3. **Publication.** The application and the international search report are published 18 months after the application filing date.
4. **Option A: Proceed to National Phase.** The applicant may now forgo Chapter II and proceed directly to the national phase and begin prosecuting the application in the countries in which protection is desired. Entry into the national phase must occur 30 months after filing of the PCT application.
5. **Option B: Chapter II.** If the applicant desires to enter Chapter II, a "demand" is filed, an international preliminary examination will be conducted, and an international preliminary report on patentability will be provided. The applicant must enter the national phase 30 months after filing of the PCT application.
6. **National Phase.** The patent offices of the countries in which patent protection is desired will now examine the application and grant or deny the patent.

EXHIBIT 21–2 The PCT Process in a Nutshell

and the costs of prosecution as the process proceeds in Japan. On the other hand, under the PCT, the applicant can file one application in the United States (which is deemed to designate all PCT countries) and later elect to prosecute the application solely in Japan. The applicant then has 20 months during Chapter I while a search is done, and, whether or not an examination is requested under Chapter II, a total of 30 months before he or she must decide whether to pursue the application by entering the national phase in Japan and pay the appropriate fees, arrange for a translation, and prosecute the application in Japan. In addition to the benefits afforded by this time, the PCT process also affords the benefits of a single application format and a centralized filing, searching, and examination system. The PCT, however, is not applicable to design patents, and therefore, the maximum delay afforded an applicant for a design patent to decide whether to make foreign patent applications is the six months provided for design patents under the Paris Convention.

The time afforded by the PCT allows an inventor additional opportunity to reflect on whether protection is actually desired in certain countries and to gather funds necessary to prosecute the applications during the national phase and to market and exploit the invention. Additionally, due to the highly authoritative nature of the international search report and written opinion (from Chapter I) and international preliminary report on patentability (from Chapter II), the PCT process provides an indication to the inventor about the likelihood of whether a patent will be granted in various countries. Finally, the time and costs involved in the national phase are likely to be significantly reduced because much of the searching and examination work required has already been conducted in Chapters I and II.

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The success of the PCT process is demonstrated by the fact that in 1979, only 2,625 PCT applications were filed, while in 2010, the number had grown to 164,300. The United States is the country of origin for approximately 27 percent of PCT applications, followed by Japan (20 percent), Germany (11 percent), China (8 percent), and the Republic of Korea (6 percent).

THE EUROPEAN PATENT ORGANIZATION

The **European Patent Organization** (EPO) was founded in 1973 to provide a uniform patent system in Europe. A European patent can be obtained by filing a single application with the EPO headquartered in Munich (or its sub-branches in The Hague or Berlin or with the national offices in the contracting nations). The application is deemed to designate all contracting states in the EPO, but the applicant must later confirm the designation for the specific countries in which protection is desired. Once granted, the patent is valid in any of the EPO countries designated and has the same force as a patent granted in any one of the contracting nations. The EPO contracting nations are Albania, Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, former Yugoslav Republic of Macedonia, France, Germany, Hellenic Republic (Greece), Hungary, Iceland, Ireland, Italy, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Monaco, Netherlands, Norway, Poland, Portugal, Romania, San Marino, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, and the United Kingdom. Although it is not a contracting nation to the European Patent Convention, Bosnia and Herzegovina has an extension agreement with the Convention, allowing it to participate as an EPO country and to be designated in an EPO patent application.

After the application is filed, a search will be conducted, the application will be examined, and the application will be published. Within six months after publication, the applicant must decide whether to pursue the application by requesting a substantive examination. It generally takes nearly four years to obtain a European patent. Within nine months following the date of grant of the patent, a party may oppose the grant on the basis the patent does not comply with the substantive provisions of the European Patent Convention. A binding decision will be issued by the opposition division of the EPO. The European patent is valid for 20 years from the date the application was filed. Since the filing of the first European patent application in 1978, more than four million European patents have been granted. In 2010, the EPO received more than 235,000 patent applications.

An applicant who is a national or resident of a nation that is a party to the European Patent Convention may file a PCT application with the EPO. A party from outside the European Patent Convention (EPC) countries who designates a contracting state in a PCT application may opt for the effect of a European patent application. Thus, if a citizen from the United States files a PCT application and desires patent protection in Germany and France, the applicant may opt for the effect of a European patent application to cover all of the European contracting nations, rather than merely later electing Germany and France, and later decide the EPC contracting states in the protection is desired.

As discussed earlier, the EPO takes a far more conservative view in granting patents for software and for business methods than does the United States.

Since 2000, there have been proposals for a unitary or single patent that would be valid across all members of the European Union. In brief, the classic bundle of European patents granted by the EPO would be transformed (upon request by the patent owner) into a unitary patent valid for all participating European Union member nations. As of the writing of this text, the proposal has gained momentum

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and many experts expect the first unitary patent to be granted in 2013. Similar proposals for a single European Patents Court have progressed, with many experts expecting the court to be approved in 2012.

THE PATENT PROSECUTION HIGHWAY

In 2006, the United States launched its **Patent Prosecution Highway** (PPH), which fast-tracks examination of corresponding patent applications filed in the USPTO and in various IP offices around the world. Under the PPH program, an applicant who receives a ruling from an "Office of First Filing" that at least one claim is patentable may request that the "Office of Second Filing" fast-track the examination of corresponding claims in corresponding applications filed in the Office of Second Filing. The Office of Second Filing can use the Office of First Filing's work products, such as search and examination reports, to streamline and expedite patent processing. For example, assume that an inventor files an application for a patent in Japan. If at least one claim is ruled patentable, the applicant may then request that the USPTO fast-track the examination of corresponding claims in a corresponding USPTO application. The USPTO will advance the application and examine it before others that may have been filed earlier. The USPTO also benefits from work previously done by the other office, in turn reducing workload and improving patent quality.

The USPTO has reported that about 90 percent of PPH cases are allowed, compared with an allowance rate of less than 50 percent for non-PPH cases. Before the PPH, patent offices around the world often duplicated each other's work. Now, although each office makes its own determination of patentability, the reusing and sharing of search and examination results streamlines the process and promotes faster patent examination.

The USPTO has PPH partnerships with 15 foreign offices, including the Japanese Patent Office and the Korean Intellectual Property Office. A complete list of PPH partnerships is available on the USPTO website.

AGREEMENT ON TRADE-RELATED ASPECTS OF INTELLECTUAL PROPERTY RIGHTS

The World Trade Organization's Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) was accepted by the United States in 1995. In addition to providing enhanced protection for trademarks (by requiring all adhering nations to allow registration of service marks as well as trademarks and providing that registrations cannot be cancelled for nonuse unless three years have passed) and copyrights (by specifying that computer programs must be protected as literary works), TRIPS also strengthens international patent law.

To comply with its obligations under TRIPS, the United States was required to amend the Patent Act in several respects. The most significant change was the revision relating to the term of a patent. Until adherence to TRIPS, the U.S. utility patent term was 17 years from the date the patent was issued. Because the term of the patent did not start until issuance, applicants in some instances delayed prosecution of their patents (sometimes called "submarine patents" because they lurked below public view) while they tested the market, gathered funds, and made plans to bring the invention to the marketplace. To harmonize U.S. law with that of most foreign nations, the term of a utility patent was changed to 20 years from the date of filing of the application, thus encouraging applicants to pursue prosecution diligently. Additionally, the publication of most patents in the United States within 18 months of filing avoids the problem of submarine patents. Another significant

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change in U.S. patent law related to determining the first to invent in priority disputes. Until TRIPS, the United States ignored evidence of inventive activity abroad, thus discriminating against foreign inventors. TRIPS requires the United States to make patent rights available without discrimination in regard to the place of invention. TRIPS is administered by the World Trade Organization, headquartered in Geneva, Switzerland.

THE PATENT LAW TREATY

Negotiations in WIPO in the latter half of the 1990s produced the Patent Law Treaty (PLT), which was adopted in June of 2000 and entered into force in April 2005. The goal of the PLT is to harmonize the formal requirements established by the individual patent offices around the world and streamline the procedures for obtaining and maintaining patents. The PLT is primarily concerned with patent formalities (rather than aspects of substantive patent law). The PLT eliminates overly burdensome requirements and establishes limits on the requirements that can be imposed by the various national patent offices throughout the world. It simplifies and standardizes application procedures that at present vary from nation to nation. For example, the PLT signatories have agreed to a set of standardized forms and have agreed that a failure to comply with various formalities at the time of filing of a patent application will not result in a loss of the filing date. Additionally, the signatories have agreed to offer electronic filing of applications and other communications. These simplified procedures make it easier and less expensive for individual inventors to apply for patents. Additionally, the PLT provides that in the event of a late filing (meaning one that is beyond a priority period), restoration is possible if it is found that the failure to file was unintentional or due care was exercised. Finally, the PLT does not require that a representative be employed during all stages of prosecution; however, a legal representative must still be engaged to provide translations, often the costliest part of the patent application process.

Only 29 countries have formally accepted the PLT, which continues the trend of harmonizing patent law throughout the world so that inventors have easier access to patent protection on an international basis. The United States has signed the PLT but has not formally ratified it. (See Appendix A for a Table of Treaties, identifying the member nations of various treaties and organizations.)

FOREIGN FILING LICENSES

To ensure that national security is not impaired, a person may not file a patent application in another country for an invention made in the United States unless the Commissioner of Patents grants a license allowing the foreign filing or until six months after the filing of the U.S. application for the invention. 35 U.S.C. § 184. The six-month waiting period allows the USPTO to review applications that might affect matters of national security.

Filing a patent application with the USPTO is deemed to be a request to the Commissioner for a license to file an application in a foreign country. The official USPTO filing receipt will indicate to the applicant whether the license is granted or denied. If the inventor does not wish to file an application in the United States but prefers to file immediately in a foreign country, he or she may file a petition to the Commissioner for Patents requesting that a **foreign filing license** be granted.

If the foreign filing license requirements are violated, any corresponding U.S. application is invalid. Additionally, criminal penalties and fines may be imposed. The violation may be cured, however, and a retroactive foreign filing license may be granted if failure to obtain the license was through error.

It is possible that the Commissioner may refuse the applicant permission to file an application in a

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foreign country and may order that the invention be kept secret. In fact, all patent applications filed in the United States are screened for subject matter that might affect national security. If the appropriate government agency determines that national security is impaired a secrecy order will be issued. A **secrecy order** prohibits publication or filing until the order is lifted. No patent can issue on an application subject to a secrecy order. The applicant may, however, obtain compensation from the government for damages caused due to his or her inability to secure a patent for the invention.

APPLICATIONS FOR U.S. PATENTS BY FOREIGN APPLICANTS

The patent laws of the United States make no discrimination with respect to citizenship of the inventor. Any inventor, regardless of his or her citizenship, may apply for a patent in the United States on the same basis as a U.S. citizen. In fact, approximately 50 percent of the patent applications received by the USPTO come from abroad. Compliance with U.S. patent law is required.

If the applicant is a citizen of a Paris Convention nation and has first filed the application in a foreign country, the applicant may claim the filing date of the earlier filed application (as long as the U.S. application is filed within 12 months after the filing of the foreign application). The U.S. application will then be treated as if it were filed on the earlier filing date.

An oath or declaration (or substitute statement, under the AIA) must be made with respect to the U.S. application. This requirement imposed on all applicants for U.S. patents is somewhat different from that of many foreign nations in that foreign nations often require neither the signature of the inventor nor an oath of inventorship.

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- Of the $12.5 million Google paid in 2011 to purchase Motorola's cell phone business, $9.5 million reportedly related to the purchase of Motorola's patents.
- In 2010, U.S. residents filed 254,895 patent applications with the USPTO. Residents of foreign nations filed 255,165 applications, with residents of Japan filing the most applications, followed by citizens of Germany, the Republic of Korea, Taiwan, Canada, the United Kingdom, and France.
- In 2011, China passed the United States and Japan to become the world's top patent filer.
- In 2010, four U.S. companies (IBM, HP, Microsoft, and Intel) were on the top-10 list of private sector patentees that received patents from the USPTO. Samsung, a South Korean company, was the number two recipient. The remaining five recipients were Japanese or South Korean companies.
- In fiscal year 2011, the USPTO received more than 48,000 international or PCT applications, representing 9 percent of all applications filed with the USPTO.
- The average total patent pendency in 2011 was 33.7 months. The average pendency for patents related to computer architecture, software, and information security was 39.6 months.
- About 2 percent of the USPTO's patent revenue is derived from PCT fees.
- In April 2011, WIPO received its two-millionth PCT application, which was filed by Qualcomm, a U.S. company.

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CHAPTER 24

Intellectual Property Audits and Due Diligence Reviews

CHAPTER OVERVIEW

Clients are often unaware of the importance of their intellectual property. To help clients realize the value of such assets and exploit them, law firms often conduct intellectual property audits for clients. The audit reveals the intellectual capital owned by a client and assists in developing a strategy so the client can maintain its valuable intellectual capital. Once clients fully understand *what* they own, they can then protect it, license it, or sell it. The audit should be repeated on a periodic basis to reflect the changing nature of intellectual property. Audits are also conducted when a company is sold, when it borrows money, or when it acquires another company. This type of audit is usually referred to as a "due diligence" review.

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INTRODUCTION

Although clients are always aware of the value of their tangible assets (such as their stock and inventory), they are often unaware that they own other valuable assets: their intellectual property. They may use distinctive names for certain products or services, may possess creative marketing materials, or may have developed a novel machine or process. All of these developments are assets that can and should be protected as intellectual property. Distinctive names should be registered as trademarks, written materials should be protected by copyright notices and registration, inventions should be protected by patents, and trade secrets should be protected so they can endure perpetually. Ocean Tomo, an IP consulting firm, has estimated that tangible assets represented about 80 percent of the value of most S & P 500 companies in the United States in 1975. By 2010, only 20 percent of these companies' market value consisted of tangible assets, with the remaining 80 percent consisting of intangible assets (including brand name, reputation, and intellectual property).

If intellectual property is not protected, it may be lost. Failure to monitor and police infringing activities may also lead to a loss of rights. Competitors may acquire rights to valuable property that formerly provided a competitive edge to a company, resulting in a loss of market share and profits.

Companies not only should protect their intellectual property in order to ensure business survival but also should use their intellectual property to create revenue. Trademarks, copyrights, patents, and trade secrets can all be licensed to others. The owner of intellectual property can achieve a continual revenue stream through licensing of rights to others for either a fixed sum or recurrent royalty payments. Alternatively, intellectual property may be sold outright to another. Intellectual property can also be used as collateral so its owner can secure a loan from a bank or other institution. Just as real property or personal property such as inventory is pledged as collateral when money is borrowed, so too can intellectual property (trademarks, copyrights, and patents) serve as collateral. Intellectual property can also be used as donations. For example, a number of companies have donated patents to universities and other nonprofit institutions. The donors then take an income tax deduction for the donation, although the Internal Revenue Service has begun subjecting such claims to increased scrutiny, and Congress has limited deductions for such donations. Recall that some companies now contribute patents (and pledge not to assert their patent rights) to the Patent Commons Project, a collaborative environment that serves as a "preserve" for patents. (See Chapter 21 for additional information about patent donations and the Patent Commons Project.)

The failure to capitalize on the value of intellectual property is generally caused by a lack of awareness of just what can be protected. Many companies believe that copyright extends only to important literary works and therefore fail to secure protection for their marketing brochures or other written materials. Similarly, companies often fail to implement measures to ensure valuable trade secrets maintain their protectability. Because clients are often unaware of the great potential and value of this property, law firms often offer their clients an **intellectual property audit** to uncover a company's protectable intellectual property. The IP audit is analogous to the accounting audit most companies conduct on an annual basis to review their financial status.

Another type of IP investigation is usually conducted when a company acquires another entity or its assets. At that time, a thorough investigation should be conducted of the intellectual property of the target company to ensure the acquiring company will obtain the benefits of what it is paying for and will not inherit infringement suits and other problems stemming from the target's failure to protect its

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intellectual property. This type of intellectual property investigation is generally called a **due diligence** review inasmuch as the acquiring company and its counsel have an obligation to duly and diligently investigate the target's assets. Due diligence reviews are also conducted when a company offers its securities for public sale inasmuch as potential investors must be informed of the offeror's assets (including its intellectual property) and any claims that may arise against the offeror. Similarly, when a company is being sold, it is generally required to identify its intellectual property and make certain representations and warranties that it owns the property being sold and there are no defects in title or pending claims involving the property. Thus, due diligence must be conducted to ensure these representations and warranties can be made with respect to a seller's intellectual property.

Audits may also be triggered by changes in the law. For example, once the Federal Circuit clarified in 1998 in *State Street* (see Chapter 21) that some business methods could qualify for patent protection, a number of companies discovered they owned such patentable business methods that required protection.

Finally, audits may identify defects in a client's intellectual property (such as a lack of an effective trade secret protection program or failure to register a trademark used by the client) so that measures may be taken to protect these valuable assets.

PRACTICAL ASPECTS OF INTELLECTUAL PROPERTY AUDITS

IP audits come in all shapes and sizes. Most importantly, the audit must fit the client and respond to its needs. If a client owns a small retail shop or provides auto repair services from only one location, the sole intellectual property may consist of the business name (which may be protectable as a trademark), any logos or designs used by the client, customer lists, and marketing materials, if any. On the other hand, if a client is engaged in software development, information technology, or telecommunications services, it may possess a wealth of IP assets. For example, in 2010 alone, IBM Corp. was granted 5,866 patents, helping to generate about $1 billion in royalties for IBM each year (according to *BusinessWeek*).

In most cases, the first IP audit is the most extensive and expensive. Some clients conduct periodic audits, and law firms typically docket the dates for annual reviews and send reminder notices to clients. The annual reviews can focus on changes since the previous audit. While there is some expense involved in conducting any audit, the benefit of the audit outweighs the expense involved. Moreover, there are often intangible benefits to the audit. If the client later wishes to obtain a loan or sell some of its assets, the preceding audit need only be updated, eliminating costly delays in a transaction. In addition, if new members join a company's IP legal team, an existing audit will help familiarize them with the company's IP assets. If the client is adequately prepared for the audit and actively assists in the audit, costs can be reduced.

Before the audit is conducted, the law firm and the client should agree on its scope and nature. Consideration should be given in regard to whether the firm will conduct the audit on an hourly fee basis or for a fixed fee. In most instances, the law firm will need to send IP professionals to the client site, resulting in disruption to the client's operations. Again, with careful preparation, such disruption can be kept to a minimum.

The following issues should be clearly addressed before the audit begins:

- Who will conduct the audit? Usually, counsel (inside or outside), together with IP paralegals, will conduct the audit, relying on company

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representatives for assistance. The law firm and the client should each designate a person who will serve as the team leader and to whom questions and concerns can be addressed. If a client is unusually large, it may designate various leaders, for example, one from its research and development department, one from its marketing department, and so forth.

- What scope will the audit have? Should only U.S. rights be explored? Should consideration be given to protecting intellectual property on an international basis?

CONDUCTING THE AUDIT

The first step in the audit should be a face-to-face meeting of the legal team and company managers. The legal team should make a brief presentation on what intellectual property is, why it is important to the company, and why and how the audit will be conducted. Managers will be more likely to cooperate if they fully understand the importance of the audit. Obtaining this kind of "buy in" from the client's managers and employees will speed the audit and reduce costs. Moreover, education about the importance of intellectual property helps ensure that managers consider ways to further protect a company's valuable assets and remain alert to possible infringements of the company's intellectual capital or infringements by the company of others' rights. Finally, having outside counsel involved in the process will ensure that confidential communications related to the audit are protected by the attorney-client privilege.

Once the company's managers have been advised of the need for the audit, the legal team should provide a worksheet or questionnaire (see Exhibit 24–1) to the company specifying the type of information that the firm is looking for so that company files can be reviewed and materials assembled for inspection by the firm and its representatives. Although it is not strictly necessary that the client do this kind of preparatory work, the more work the client does, the faster, cheaper, and less disruptive the audit will be.

Once the materials that are responsive to the questionnaire are gathered, the legal team can review them. The review is generally done at the client site. Files are pulled, brochures gathered, and contracts assembled. The legal team will review these materials and often make copies of pertinent documents, marking them as confidential.

After review of the materials is completed, another face-to-face meeting should be held to ensure that all of the materials were gathered and that there are no other contracts, license agreements, or other documents or software that should be reviewed. Questions may have arisen during the course of the audit for which the legal team needs responses. For example, the legal team may ask whether a certain brochure or trademark is still in use or whether the company logo was designed in-house or by an independent contractor. The legal team will generally check the records of the USPTO and the Copyright Office to determine whether there are any records on file showing the company's ownership of trademarks, copyrights, and patents. A follow-up questionnaire may be sent to the client to obtain the answers to questions that arose in the course of the audit.

POSTAUDIT ACTIVITY

After the inspection is completed, the legal team will usually prepare a written report identifying the specific items of IP owned by the company, reviewing their status, and making recommendations for protection. If outside counsel prepares the report, confidential information in the report will be protected by the attorney-client privilege, such that it will not be discoverable. The IP audit team may then proceed to take the following actions: filing applications

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Intellectual Property Audit Questionnaire

Identified below is a list of subjects to be covered in connection with our review of the intellectual property rights of your company (the "Company"). Please gather any materials and documents relevant to the subjects listed below so they can be reviewed. If no materials or documents exist with respect to a particular subject, please confirm that in writing next to the relevant question.

For all relevant questions, give the name of the person who assisted in the design, development, or implementation of any intellectual property rights; describe the nature of the relationship between the person and the Company (for example, employee, independent contractor, and so forth); and indicate whether the person signed any confidentiality or nondisclosure agreements with the Company.

A. General

1. Has the Company ever acquired another entity or the business or assets of another entity or person? (This party may have owned intellectual property that is now owned by the Company.)
2. Has the Company ever sold any assets or business to another entity or person?
3. Is the Company engaged in the development or design of any useful products or parts for useful products?
4. Is the Company engaged in the development, design, or modification of computer software?
5. Are there any individuals who may have developed or designed products or software for the Company who have left employment with the Company? If so, indicate whether each such person signed a confidentiality, nondisclosure, or employment agreement with the Company.
6. Does the Company have a website or are there links from other parties' sites to the Company's site? If so, provide documents relating to the registration of the Company's domain names and website addresses.
7. Does the Company use a docketing system or calendar to provide reminders of due dates relating to any of its intellectual property?
8. Has the Company ever conducted an intellectual property audit, or does it maintain a list of its IP assets? If so, please attach a copy.
9. Has the Company changed its name or address within the past 10 years? If so, records on file with the U.S. Patent and Trademark Office (USPTO) and Copyright Office may need to be updated to reflect such.

B. Trademarks and Service Marks

1. Does the Company use any trademarks, service marks, logos, slogans, trade dress, or designs (collectively "Marks") in connection with the offer and sale of its products or services? These Marks may have been displayed on products, labels, packaging, letterhead, business cards, in advertisements, brochures, or other marketing materials, including a website.
2. Has the Company ever applied for registration of any Marks with the USPTO or any state trademark agencies?
3. Has the Company ever allowed another party (such as an employee, client, vendor, or competitor) to use any of the Company's Marks?
4. Does the Company use any Marks that are owned by any third party? Review Company marketing materials, website, and other written documents to determine whether the Company displays or uses Marks belonging to another.

C. Copyrights

1. What written materials does the Company use to advertise its products and services? These may be written materials, scripts or copy for radio or television advertisements, or website materials.

EXHIBIT 24–1 Intellectual Property Audit Questionnaire

(*Continues*)

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Were these developed by Company employees or by independent contractors working for the Company? What agreements exist relating to the development of such materials?

2. What written materials does the Company use internally, such as employee handbooks, training materials, company policies, manuals, training or other videos, audios, or modules, relating to the way the Company conducts its business?
3. Do Company employees prepare written materials or electronic materials (such as PowerPoint slides) when presentations are made within the Company and outside the Company? If so, describe.
4. Do Company employees submit articles for publication to any journals, periodicals, or other publications? If so, describe.
5. Has the Company ever applied for registration of any copyrights with the U.S. Copyright Office?
6. Are articles from periodicals, magazines, trade journals, and other related written materials photocopied for distribution within the Company?
7. What policies exist regarding reproduction of books, articles, journals, and other materials that may be subject to copyright?
8. Is any music piped in through the Company's offices (whether through the use of CDs or music being simultaneously transmitted by a radio station)?
9. Is music played when callers to the Company are placed on hold?

D. Patents

1. Has the Company or its employees ever invented any useful article, product, method, process, or software? If so, do any inventor or laboratory notebooks exist that document the development of the invention? Does the Company have any present plans for any such inventions?
2. Has the Company ever made any improvements to another party's useful article, product, or invention?
3. Has the Company ever engaged in the reverse engineering or decompilation of another company's product or software?
4. Has the Company ever applied with the USPTO for issuance of any patent?
5. Does the Company mark all patented inventions with a patent notice?
6. What process does the Company use to ensure all notices of patent are removed when the patent has expired?

E. Trade Secrets

1. What information does the Company possess that would be harmful to the Company if it were discovered by a competitor? Consider proprietary information relating to research and development plans; calculations and financial data; employee manuals and handbooks and personnel information; information relating to the Company's clients and customers; the Company's methods of recruiting; methods and processes of production; test results for Company products and services; data concerning the pricing for Company products and services; sales forecasts; research information; manufacturing information; marketing materials; surveys and data relating to customer needs and preferences; and business plans and forecasts.
2. Do employment, confidentiality, nondisclosure, or other agreements exist that protect such proprietary materials?
3. Does the Company conduct employee orientation for new employees and exit interviews for departing employees to ensure employees understand the need to maintain confidentiality of Company trade secrets?

EXHIBIT 24–1 (*Continued*)

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4. What measures does the Company take to protect confidential and proprietary information? Are documents marked “Confidential”? Are proprietary materials kept in locked cabinets? Are restrictions placed on the access to and copying of such materials? Is access to Company premises monitored by cameras, key access, or any other method?
5. What measures does the Company take to ensure electronic communications are protected and secure? Are any encryption methods in place for electronic communications? Describe restrictions on access to the Company’s computer systems, including password protection methods.
6. Are any legal notices or disclaimers provided at sign-on when Company employees access the Company network?

F. Software

1. Has the Company designed, developed, or modified any software?
2. Does the Company have the right to use any software designed or developed by another party? If so, have licenses been obtained for each user?
3. Does the Company allow its employees to copy any software for their home use?

G. Claims

1. Are there any presently pending claims relating to any of the Company’s intellectual property? Has any person alleged or claimed that the Company has violated or infringed its intellectual property rights?
2. What claims have been made (whether or not resolved or compromised) within the past five years against the Company relating to alleged infringement or violation by the Company of the intellectual property rights of others?
3. Has the Company observed or made any claims relating to any possible infringements by others of the Company’s intellectual property rights?
4. Does the Company use any methods to detect possible infringing uses of its intellectual property?

H. General Documents

1. Please assemble all applications and registrations, and all license, royalty, security agreements or other agreements relating to trade names, Marks, copyrights, patents, trade secrets, software, licenses, or other similar rights relating to the Company’s intellectual property rights.
2. Please assemble all agreements entered into by employees (including officers and directors) and any Company handbooks, manuals, or policies relating to Company employees.
3. Please assemble all agreements entered into between the Company and any consultants or independent contractors.
4. To the extent not already covered, please assemble all agreements or licenses entered into between the Company and any other parties that relate to the Company’s intellectual property, which are essential for the operation of the Company’s business, or by which the Company is allowed to offer or use another party’s Marks, copyrights, patents, processes, software, or other related property, including contracts entered into with the government.
5. Does the Company offer its products or services in any foreign countries? If not, does the Company have any plans to offer its products or services in foreign countries within the next three years?

EXHIBIT 24–1 (*Continued*)

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ETHICS EDGE

CONFIDENTIALITY AND COMPETENCE

Throughout this text, many of the ethics tips have focused on two critical duties of paralegals: the duty to maintain confidential information and the duty to provide competent service to clients. Those two core duties play a significant part in IP audits. Consider the following:

- IP audits routinely disclose the most important assets that many companies own. Be scrupulous in maintaining the confidentiality of the information to which you will gain access. Do not discuss the results of the audit with others, keep the information securely protected, and mark all documents with notices of confidentiality.
- As you work on clients' IP matters, maintain and update your own IP audit list. For example, when a client secures a trademark registration, note the particulars on your own client audit sheet. When it comes time to conduct a full-fledged IP audit or to update a prior audit, you will already have much of the information at your fingertips, saving the client time and money.

for registration of trademarks, service marks, copyrights, and patents; ensuring that the chain of title for trademarks, copyrights, and patents is accurate (and correcting and filing documents at the USPTO and Copyright Office if the company has changed its name or acquired another's IP); drafting contracts to be used when the company retains independent contractors (to ensure that all work created by the independent contractors is "work made for hire" or is owned by or assigned to the company); preparing license agreements so the company can license its intellectual property to others; preparing nondisclosure and noncompetition agreements; and preparing policies for the protection of trade secrets.

The IP audit team may also assist the client's human resources department and provide instructions on conducting exit interviews and may redraft the employee manual to include a trade secret policy, Internet use policy, and instructions regarding use of the company's trademarks. The legal team may also advise the company to engage appraisers and analysts to valuate the IP assets so they can be licensed at appropriate royalties or sold at their fair market value to produce revenue for the company.

The company is usually advised to initiate an aggressive campaign to locate others who may be infringing the company's intellectual property by reviewing competitors' materials and trade publications and by monitoring applications filed at the USPTO for marks that may be confusingly similar to those owned by the company and published patent applications for inventions that might infringe the company's patents. The legal team may help the company initiate a docketing system for reminder dates for renewals of licenses, trademark registrations, and patent maintenance, or may be retained to perform the docketing functions itself.

Once companies know what IP assets they own, they will be able to protect these valuable assets and mine them to produce a revenue stream by licensing or using the assets as collateral to obtain financing.

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TRIVIA

- IBM's worldwide patent portfolio includes 40,000 patents.
- *BusinessWeek* has reported that Pfizer has relied on a single set of patents covering the drug Lipitor for one-fourth of its total sales. The effect on Pfizer of the expiration of the patent for Lipitor in November 2011 is unknown.
- Many companies, such as Hewlett-Packard, post notices on their websites indicating that their IP is for sale or licensing.
- Dow Chemical reportedly used the results of an IP audit to increase its patent licensing by $125 million.
- According to the *Hollywood Reporter*, the overall revenue achieved by the licensing of Star Wars memorabilia, action figures, spin-offs, and so forth in 2010 alone was approximately $510 million.
- Starbucks derives revenue from licensing its name to Jim Beam for a coffee-flavored liqueur.
- The COCA-COLA® trademark is the most recognized trademark in the world, and the company values its brands at more than $70 billion.

CHAPTER SUMMARY

Because clients may be unaware of the value of the intellectual property they own, law firms often conduct intellectual property audits for clients. The audit reveals valuable intellectual property assets that can then be exploited for the client's benefit. Audits should be conducted on a periodic basis to reflect the changing nature of intellectual property. Audits or reviews are also conducted when companies are sold, when they borrow money, or when they acquire other companies. In such instances, the review is often called "due diligence." Because almost all types of intellectual property can be lost through lack of protection (including nonuse, failure to monitor licensees properly, failure to renew or maintain registrations, and failure to protect against infringing activities), the intellectual property audit is a crucial tool that allows a company to understand and exploit the value of its intellectual property portfolio.

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